

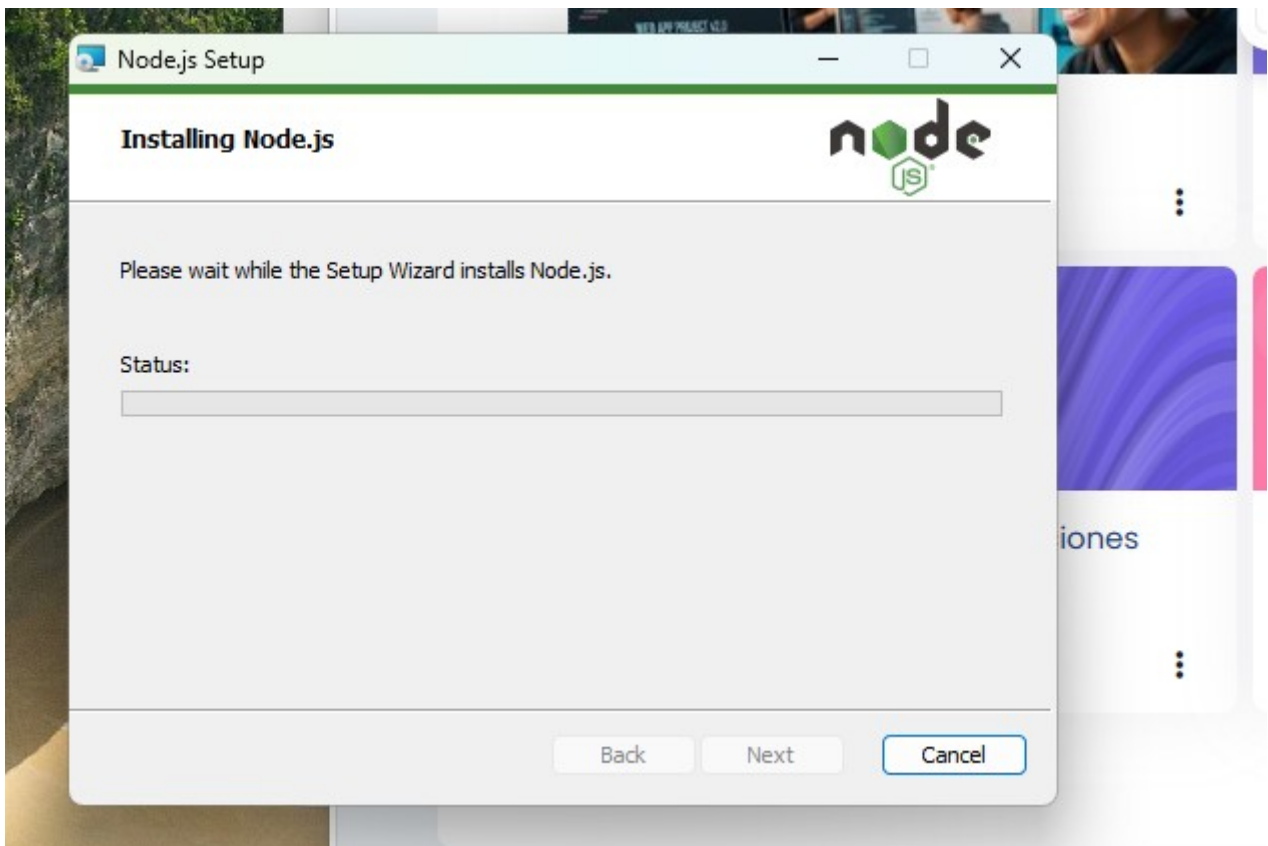
Día 18-09-2026

Daniel Copete Jiménez

Node.js es un entorno que permite ejecutar JavaScript fuera del navegador y se utiliza para desarrollar servidores y APIs.

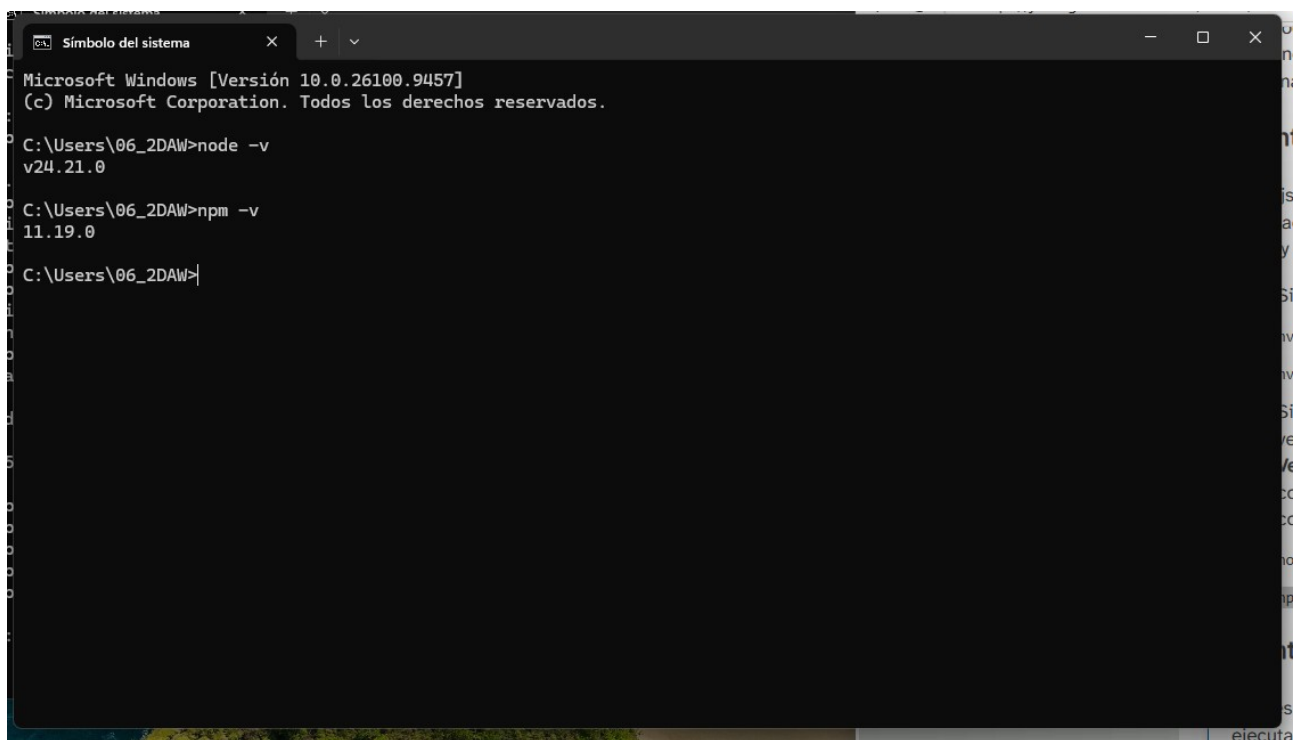
Ionic es un framework que permite crear aplicaciones móviles y web utilizando tecnologías como HTML, CSS y JavaScript. Ambas herramientas facilitan el desarrollo de aplicaciones modernas y multiplataforma.

Captura 1



En esta captura empieza la descarga de node.js

Captura 2



```
Símbolo del sistema
Microsoft Windows [Versión 10.0.26100.9457]
(c) Microsoft Corporation. Todos los derechos reservados.

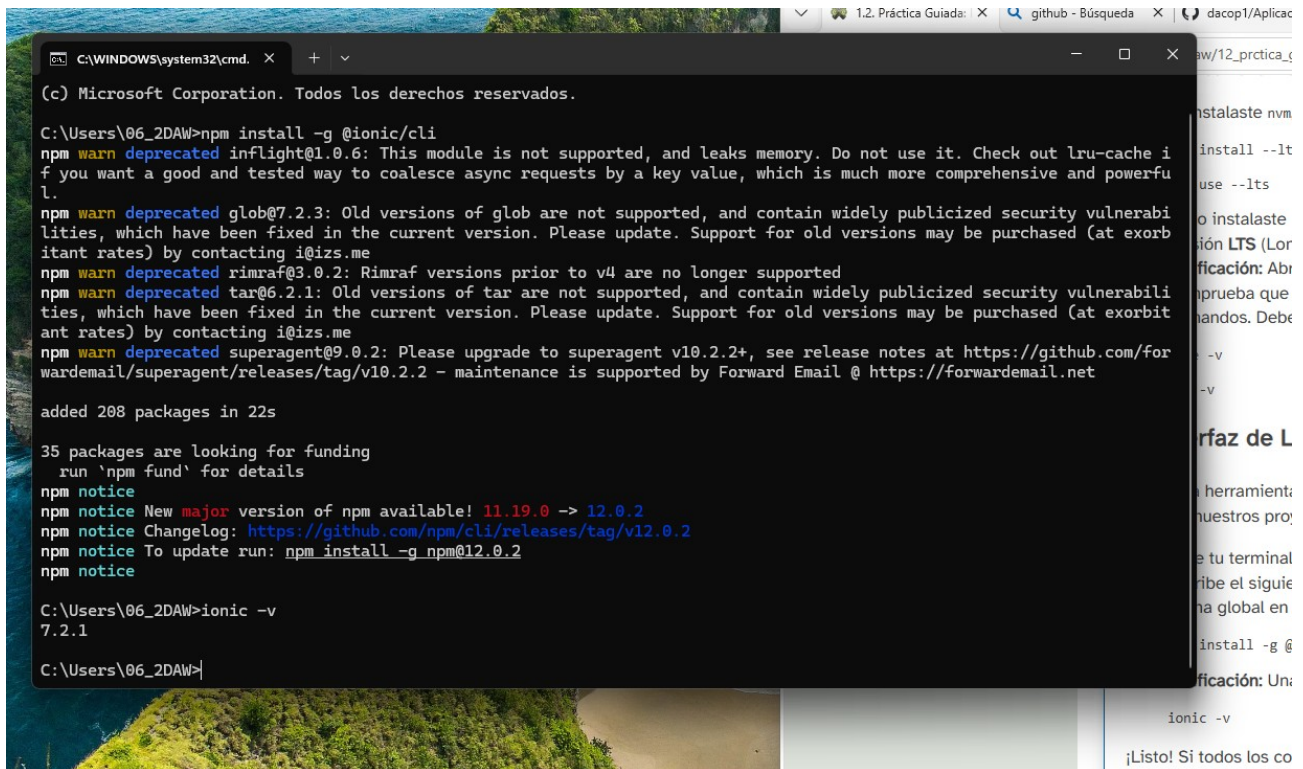
C:\Users\06_2DAW>node -v
v24.21.0

C:\Users\06_2DAW>npm -v
11.19.0

C:\Users\06_2DAW>
```

Verificamos que la instalación se ha completado de manera exitosa y verificamos las versiones son las más actualizadas

Captura 3



```
(c) Microsoft Corporation. Todos los derechos reservados.

C:\Users\06_2DAW>npm install -g @ionic/cli
npm warn deprecated inflight@1.0.6: This module is not supported, and leaks memory. Do not use it. Check out lru-cache if you want a good and tested way to coalesce async requests by a key value, which is much more comprehensive and powerful.
npm warn deprecated glob@7.2.3: Old versions of glob are not supported, and contain widely publicized security vulnerabilities, which have been fixed in the current version. Please update. Support for old versions may be purchased (at exorbitant rates) by contacting i@izs.me
npm warn deprecated rimraf@3.0.2: Rimraf versions prior to v4 are no longer supported
npm warn deprecated tar@6.2.1: Old versions of tar are not supported, and contain widely publicized security vulnerabilities, which have been fixed in the current version. Please update. Support for old versions may be purchased (at exorbitant rates) by contacting i@izs.me
npm warn deprecated superagent@9.0.2: Please upgrade to superagent v10.2.2+, see release notes at https://github.com/forwardemail/superagent/releases/tag/v10.2.2 - maintenance is supported by Forward Email @ https://forwardemail.net

added 208 packages in 22s

35 packages are looking for funding
  run `npm fund` for details
npm notice
npm notice New major version of npm available! 11.19.0 -> 12.0.2
npm notice Changelog: https://github.com/npm/cli/releases/tag/v12.0.2
npm notice To update run: npm install -g npm@12.0.2
npm notice

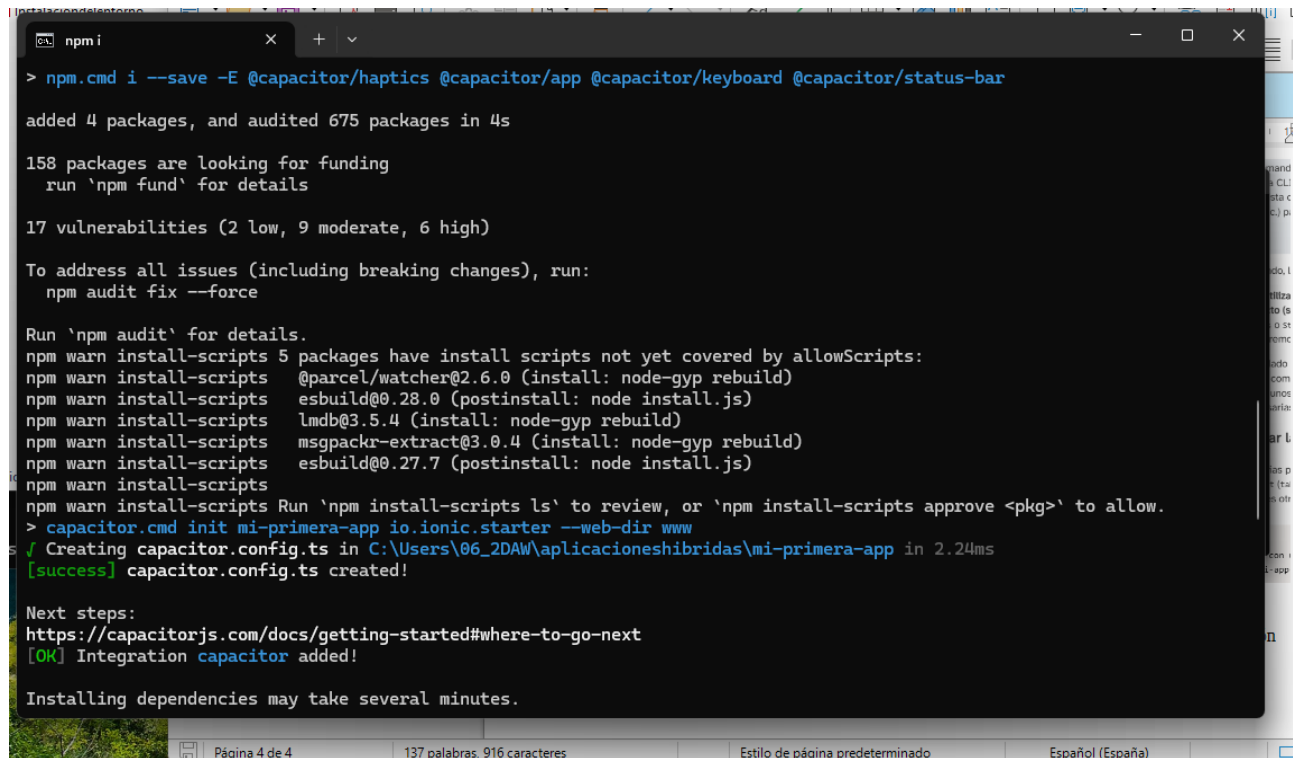
C:\Users\06_2DAW>ionic -v
7.2.1

C:\Users\06_2DAW>
```

Instalación exitosa de ionic y verificamos su versión.

Ahora vamos a crear nuestra primera aplicación

Captura04



```
npm i
> npm.cmd i --save -E @capacitor/haptics @capacitor/app @capacitor/keyboard @capacitor/status-bar
added 4 packages, and audited 675 packages in 4s

158 packages are looking for funding
  run `npm fund` for details

17 vulnerabilities (2 low, 9 moderate, 6 high)

To address all issues (including breaking changes), run:
  npm audit fix --force

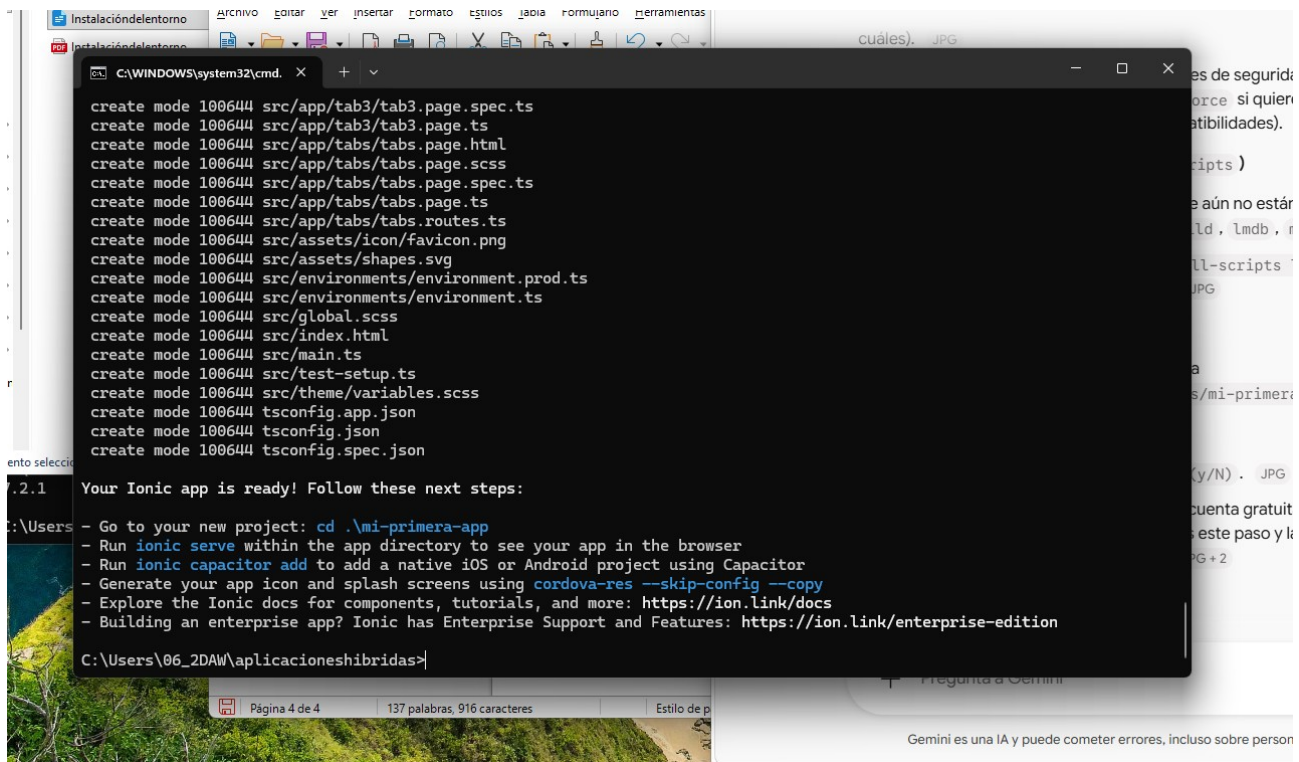
Run `npm audit` for details.
npm warn install-scripts 5 packages have install scripts not yet covered by allowScripts:
npm warn install-scripts @parcel/watcher@2.6.0 (install: node-gyp rebuild)
npm warn install-scripts esbuild@0.28.0 (postinstall: node install.js)
npm warn install-scripts lmd@3.5.4 (install: node-gyp rebuild)
npm warn install-scripts msgpackr-extract@3.0.4 (install: node-gyp rebuild)
npm warn install-scripts esbuild@0.27.7 (postinstall: node install.js)
npm warn install-scripts Run `npm install-scripts ls` to review, or `npm install-scripts approve <pkg>` to allow.
> capacitor.cmd init mi-primer-app io.ionic.starter --web-dir www
✓ Creating capacitor.config.ts in C:\Users\06_2DAW\aplicacioneshibridas\mi-primer-app in 2.24ms
[success] capacitor.config.ts created!

Next steps:
https://capacitorjs.com/docs/getting-started#where-to-go-next
[OK] Integration capacitor added!

Installing dependencies may take several minutes.
```

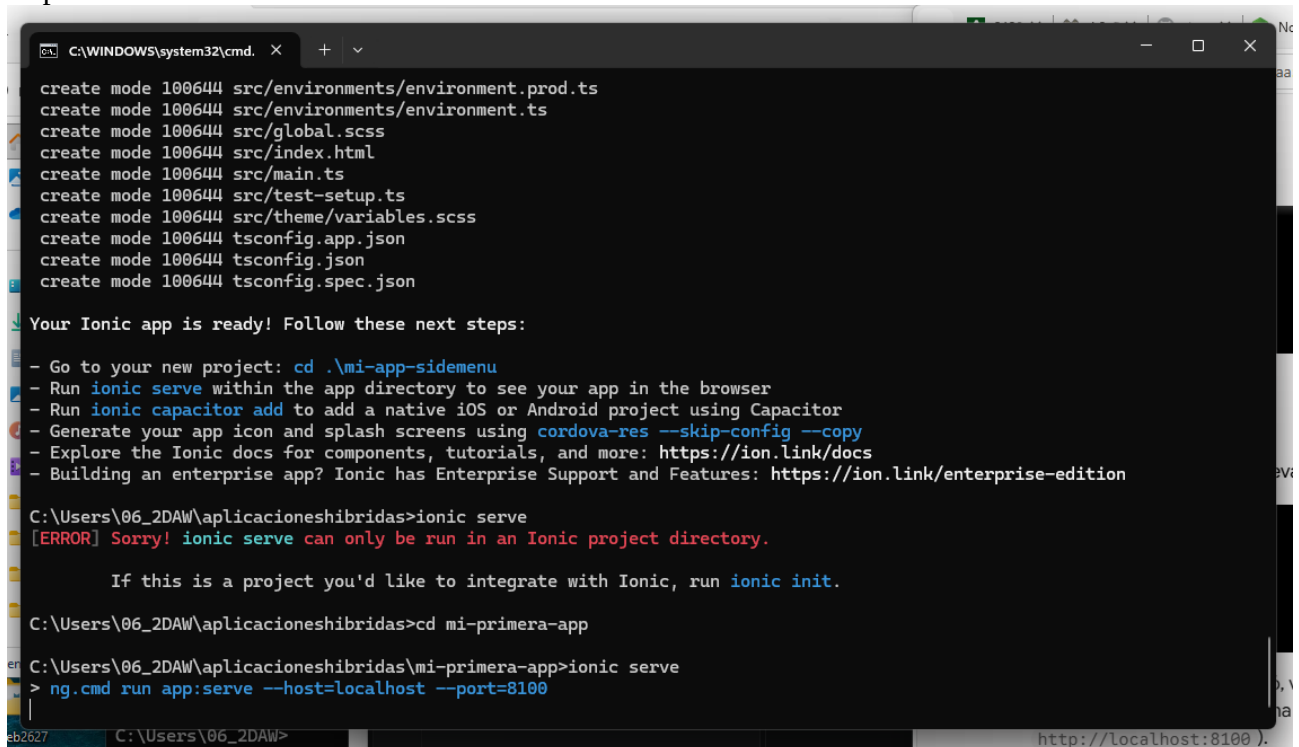
He creado una carpeta llamada aplicacioneshibridas y he accedido a ella mediante la terminal. Después, he ejecutado el comando ionic start mi-primer-app tabs para crear una nueva aplicación Ionic utilizando Angular y el diseño de pestañas (tabs).

Captura05



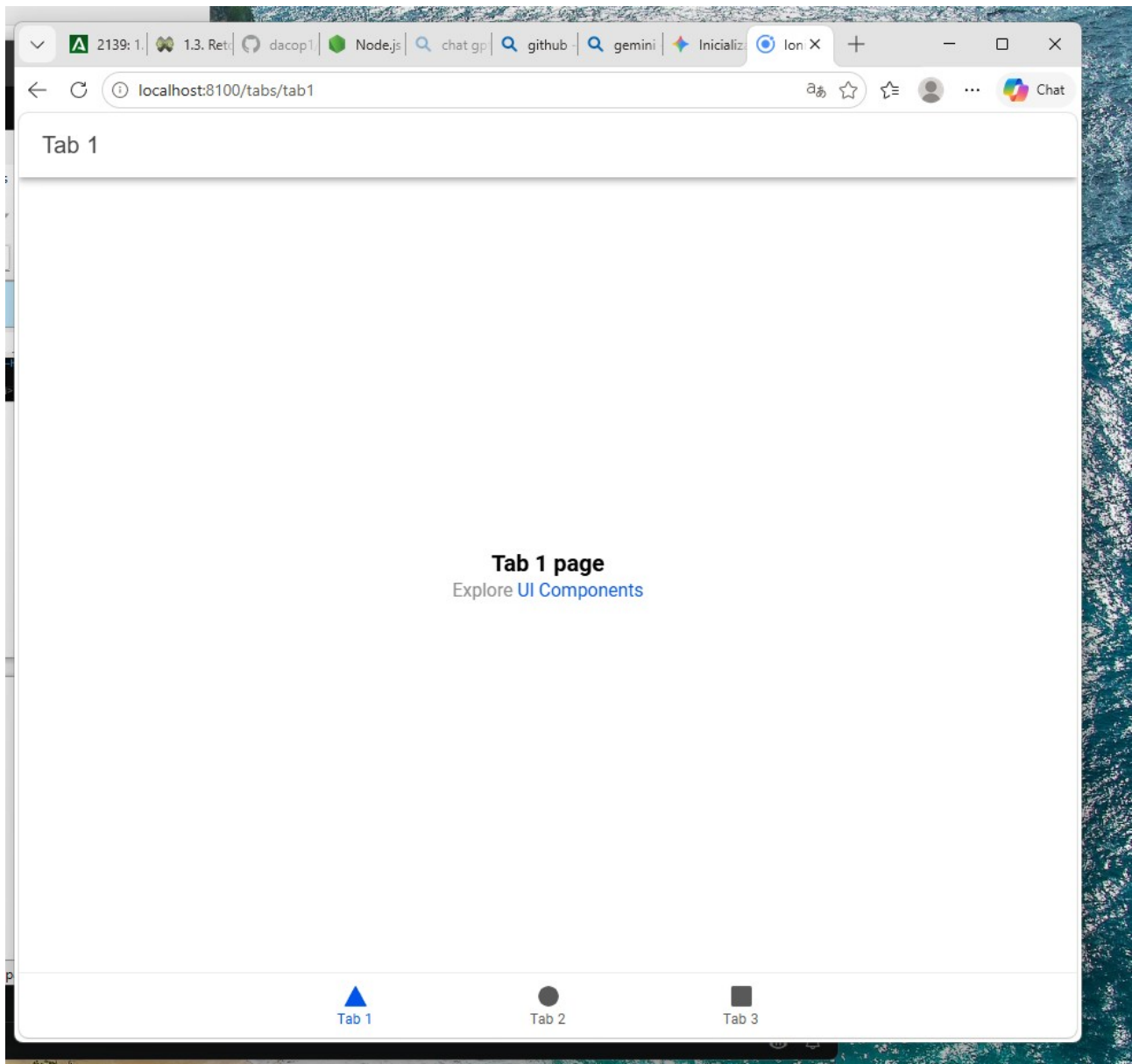
En esta captura termina la creación de la app: la terminal confirma que el proyecto está listo ("Your Ionic app is ready!") y sugiere los siguientes pasos (ionic serve, ionic capacitor add, etc.).

Captura 06



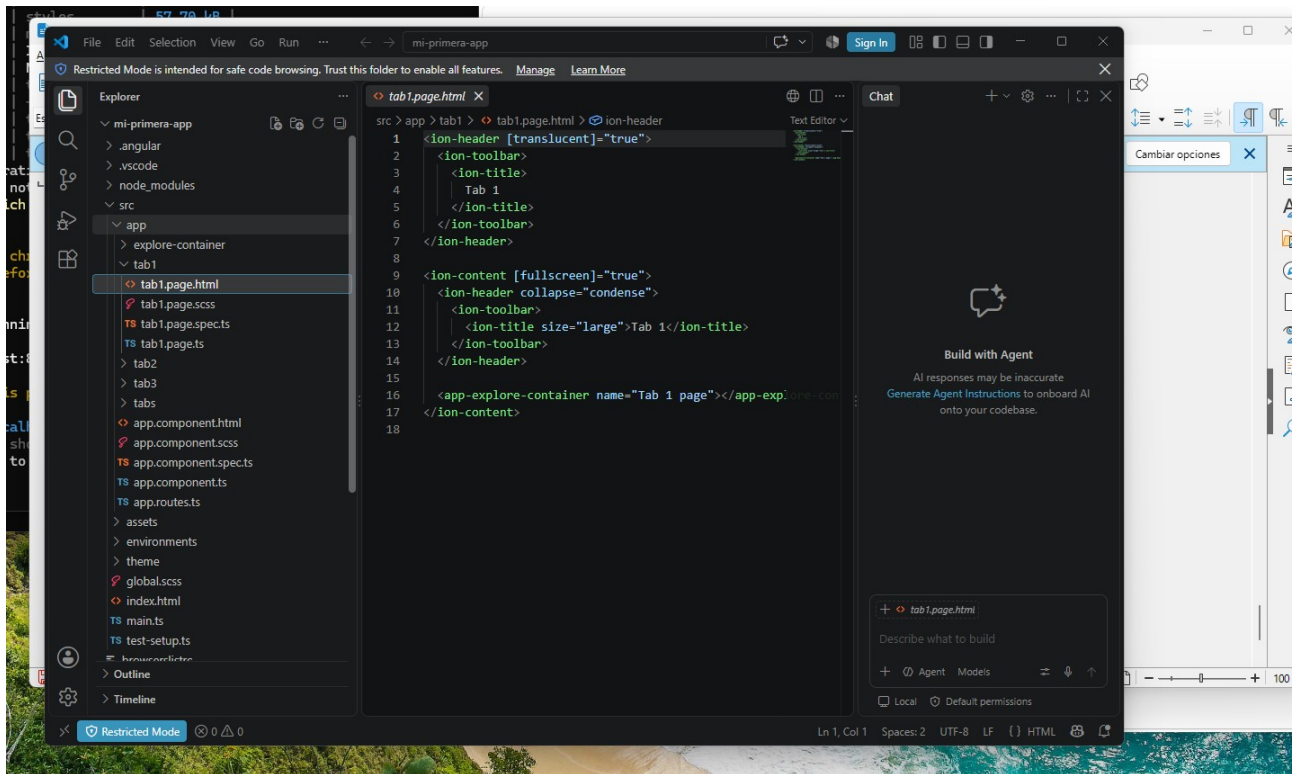
Aquí se crea también el proyecto "mi-app-sidemenu". Al ejecutar ionic serve fuera de la carpeta del proyecto da un error, y tras entrar en la carpeta mi-primer-app el servidor arranca correctamente.

Captura 07



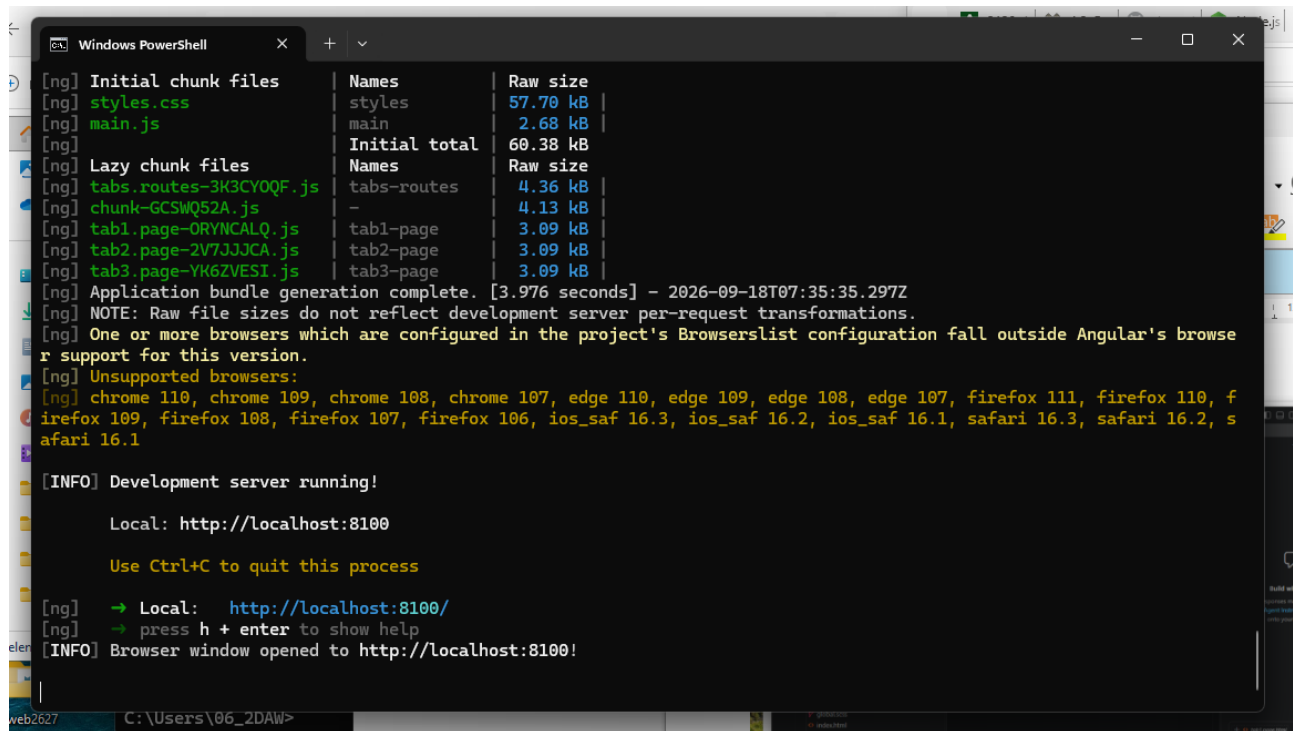
La app "tabs" ejecutándose en el navegador (localhost:8100/tabs/tab1), con la barra inferior de pestañas Tab 1 / Tab 2 / Tab 3.

Captura 09



Proyecto abierto en Visual Studio Code, mostrando la estructura de carpetas (tab1, tab2, tab3, tabs) y el archivo tab1.page.html.

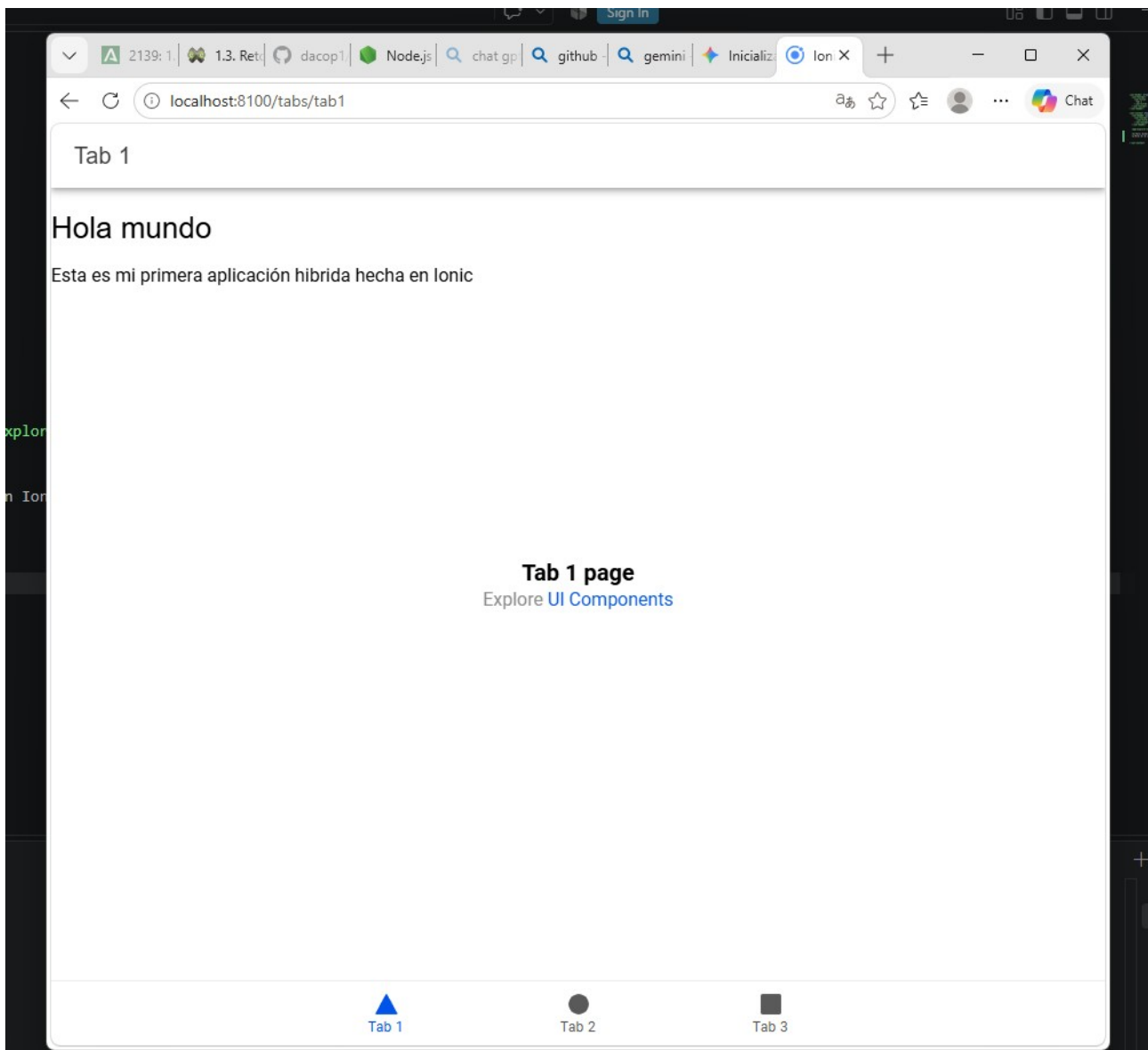
Captura 10



```
[ng] Initial chunk files      Names      Raw size
[ng] styles.css             styles      57.70 kB
[ng] main.js                main        2.68 kB
[ng]                        Initial total 60.38 kB
[ng] Lazy chunk files       Names      Raw size
[ng] tabs.routes-3K3CY0QF.js tabs-routes 4.36 kB
[ng] chunk-GCSWQ52A.js      -          4.13 kB
[ng] tab1.page-ORYNCALQ.js   tab1-page   3.09 kB
[ng] tab2.page-2V7JJJCA.js   tab2-page   3.09 kB
[ng] tab3.page-YK6ZVESI.js   tab3-page   3.09 kB
[ng] Application bundle generation complete. [3.976 seconds] - 2026-09-18T07:35:35.297Z
[ng] NOTE: Raw file sizes do not reflect development server per-request transformations.
[ng] One or more browsers which are configured in the project's Browserslist configuration fall outside Angular's browser support for this version.
[ng] Unsupported browsers:
[ng] chrome 110, chrome 109, chrome 108, chrome 107, edge 110, edge 109, edge 108, edge 107, firefox 111, firefox 110, firefox 109, firefox 108, firefox 107, firefox 106, ios_saf 16.3, ios_saf 16.2, ios_saf 16.1, safari 16.3, safari 16.2, safari 16.1
[INFO] Development server running!
      Local: http://localhost:8100
      Use Ctrl+C to quit this process
[ng] → Local: http://localhost:8100/
[ng] → press h + enter to show help
[INFO] Browser window opened to http://localhost:8100!
```

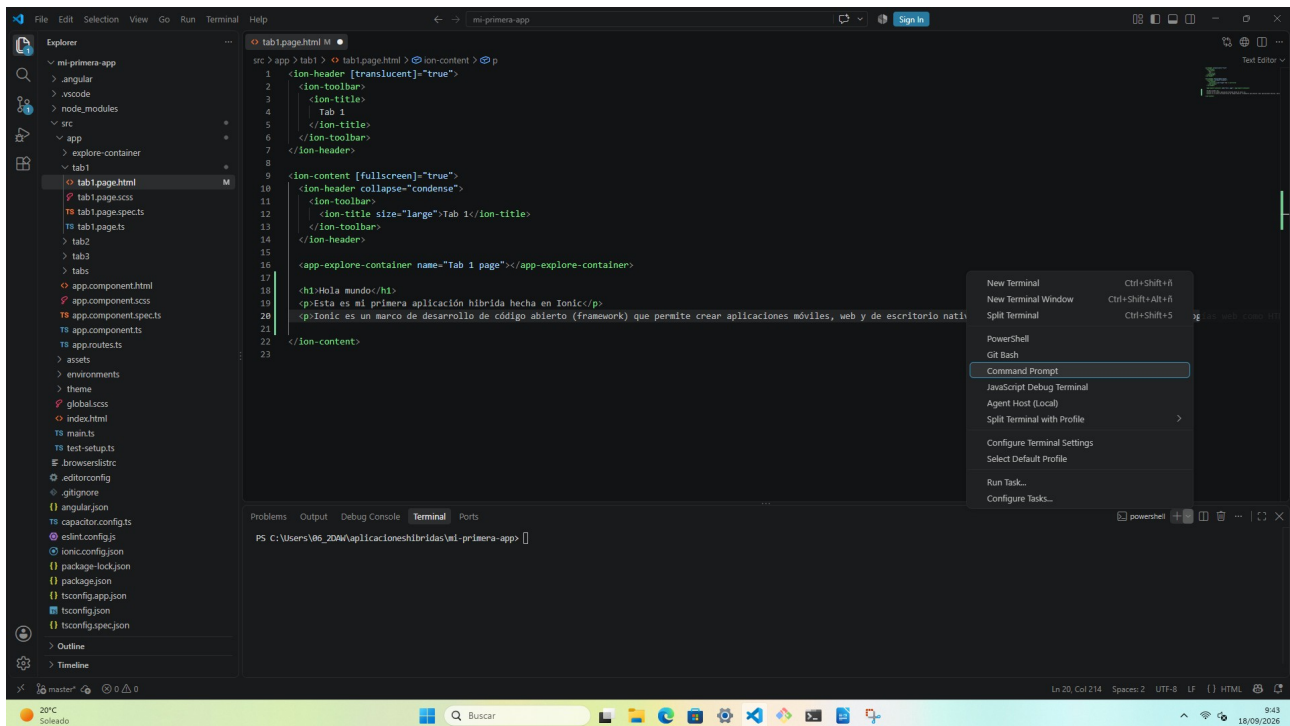
La terminal muestra la compilación de Angular y confirma que el servidor de desarrollo está corriendo en <http://localhost:8100>.

Captura 11



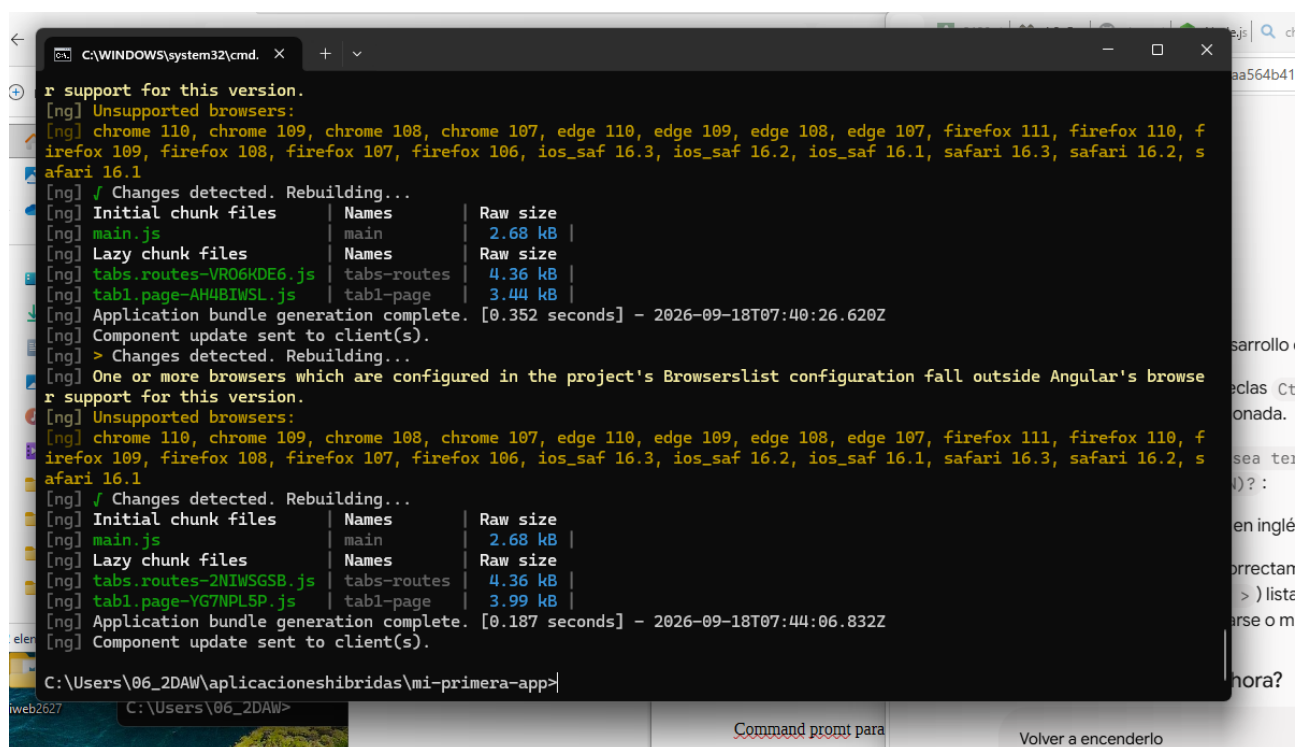
Resultado en el navegador tras editar el código de Tab 1: aparece el texto personalizado "Hola mundo, esta es mi primera aplicación híbrida hecha en Ionic".

Captura 12



Command prompt para cambiar cmd

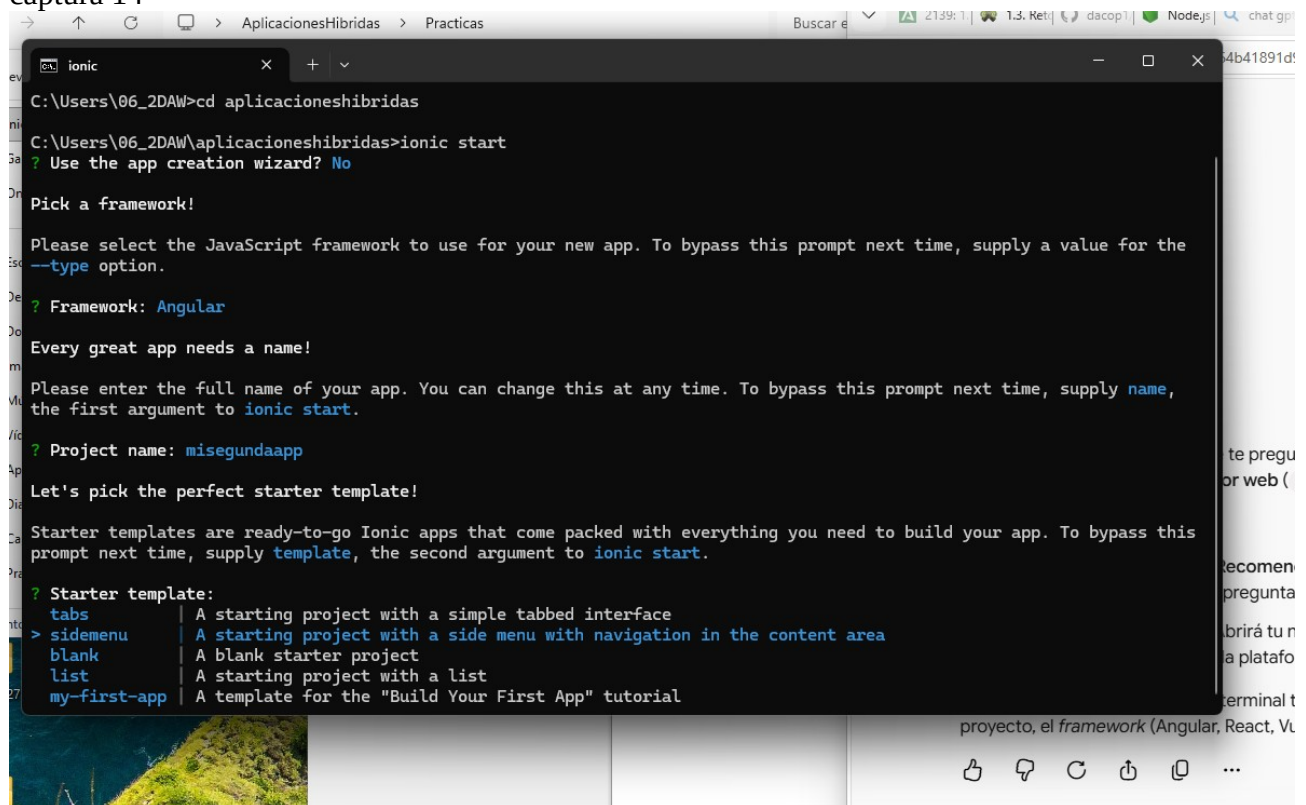
Captura 13



```
C:\WINDOWS\system32\cmd. X + v
[ng] r support for this version.
[ng] Unsupported browsers:
[ng] chrome 110, chrome 109, chrome 108, chrome 107, edge 110, edge 109, edge 108, edge 107, firefox 111, firefox 110, f
[ng] irefox 109, firefox 108, firefox 107, firefox 106, ios_saf 16.3, ios_saf 16.2, ios_saf 16.1, safari 16.3, safari 16.2, s
[ng] afari 16.1
[ng] ✓ Changes detected. Rebuilding...
[ng] Initial chunk files | Names | Raw size
[ng] main.js | main | 2.68 kB |
[ng] Lazy chunk files | Names | Raw size
[ng] tabs.routes-VRO6KDE6.js | tabs-routes | 4.36 kB |
[ng] tab1.page-AH4BIWSL.js | tab1-page | 3.44 kB |
[ng] Application bundle generation complete. [0.352 seconds] - 2026-09-18T07:40:26.620Z
[ng] Component update sent to client(s).
[ng] > Changes detected. Rebuilding...
[ng] One or more browsers which are configured in the project's Browserslist configuration fall outside Angular's browse
[ng] r support for this version.
[ng] Unsupported browsers:
[ng] chrome 110, chrome 109, chrome 108, chrome 107, edge 110, edge 109, edge 108, edge 107, firefox 111, firefox 110, f
[ng] irefox 109, firefox 108, firefox 107, firefox 106, ios_saf 16.3, ios_saf 16.2, ios_saf 16.1, safari 16.3, safari 16.2, s
[ng] afari 16.1
[ng] ✓ Changes detected. Rebuilding...
[ng] Initial chunk files | Names | Raw size
[ng] main.js | main | 2.68 kB |
[ng] Lazy chunk files | Names | Raw size
[ng] tabs.routes-2NIWSGSB.js | tabs-routes | 4.36 kB |
[ng] tab1.page-YG7NPL5P.js | tab1-page | 3.99 kB |
[ng] Application bundle generation complete. [0.187 seconds] - 2026-09-18T07:44:06.832Z
[ng] Component update sent to client(s).
C:\Users\06_2DAW\aplicacioneshibridas\mi-primera-app>
```

La terminal detecta los cambios guardados en el código y recompila automáticamente la aplicación .

mi segunda app usando sidemenu
captura 14

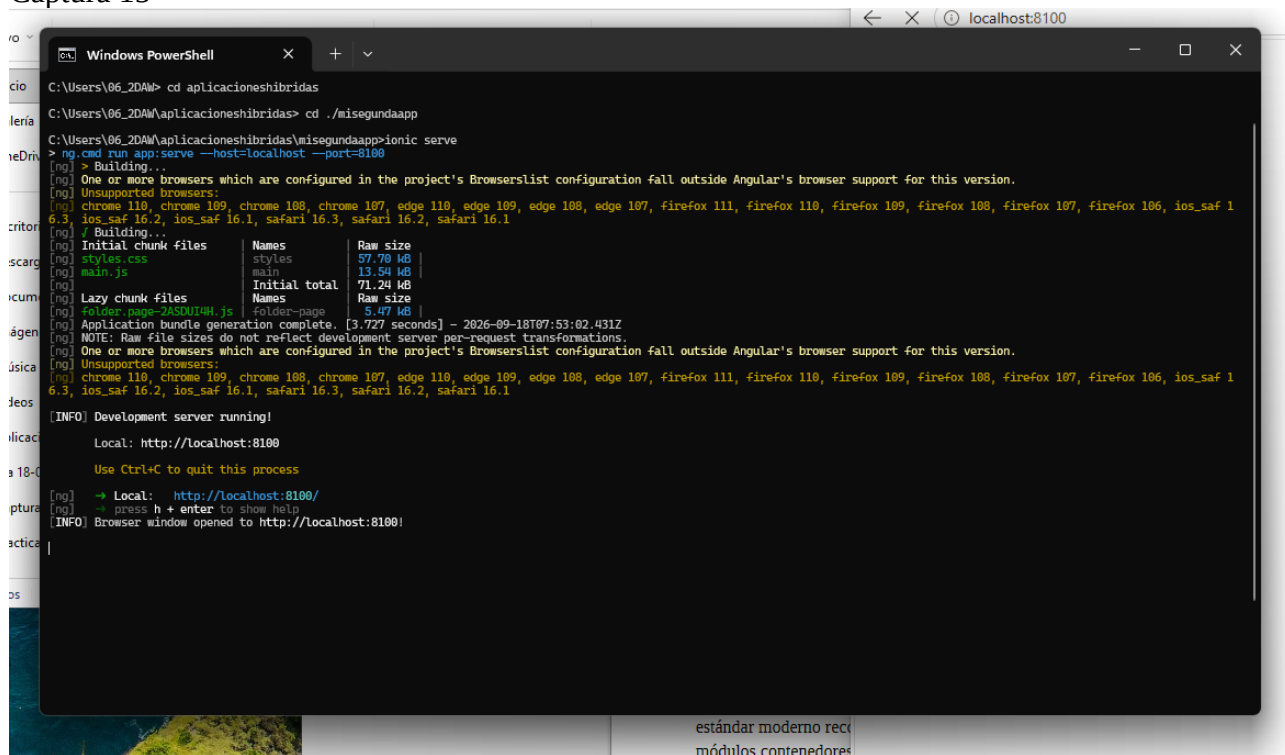


```
C:\Users\06_2DAW>cd aplicacioneshibridas
C:\Users\06_2DAW\aplicacioneshibridas>ionic start
? Use the app creation wizard? No
Pick a framework!
Please select the JavaScript framework to use for your new app. To bypass this prompt next time, supply a value for the
--type option.
? Framework: Angular
Every great app needs a name!
Please enter the full name of your app. You can change this at any time. To bypass this prompt next time, supply name,
the first argument to ionic start.
? Project name: misegundaapp
Let's pick the perfect starter template!
Starter templates are ready-to-go Ionic apps that come packed with everything you need to build your app. To bypass this
prompt next time, supply template, the second argument to ionic start.
? Starter template:
  tabs      | A starting project with a simple tabbed interface
> sidemenu  | A starting project with a side menu with navigation in the content area
  blank     | A blank starter project
  list      | A starting project with a list
  my-first-app | A template for the "Build Your First App" tutorial
```

* La diferencia principal entre **Standalone Components** y **NgModule** en Angular (el framework más usado con Ionic) radica en cómo se estructuran y gestionan las dependencias dentro de la aplicación.

NgModule es el enfoque tradicional (utilizado desde Angular 2), mientras que **Standalone** es el estándar moderno recomendado que elimina la necesidad de usar módulos contenedores.

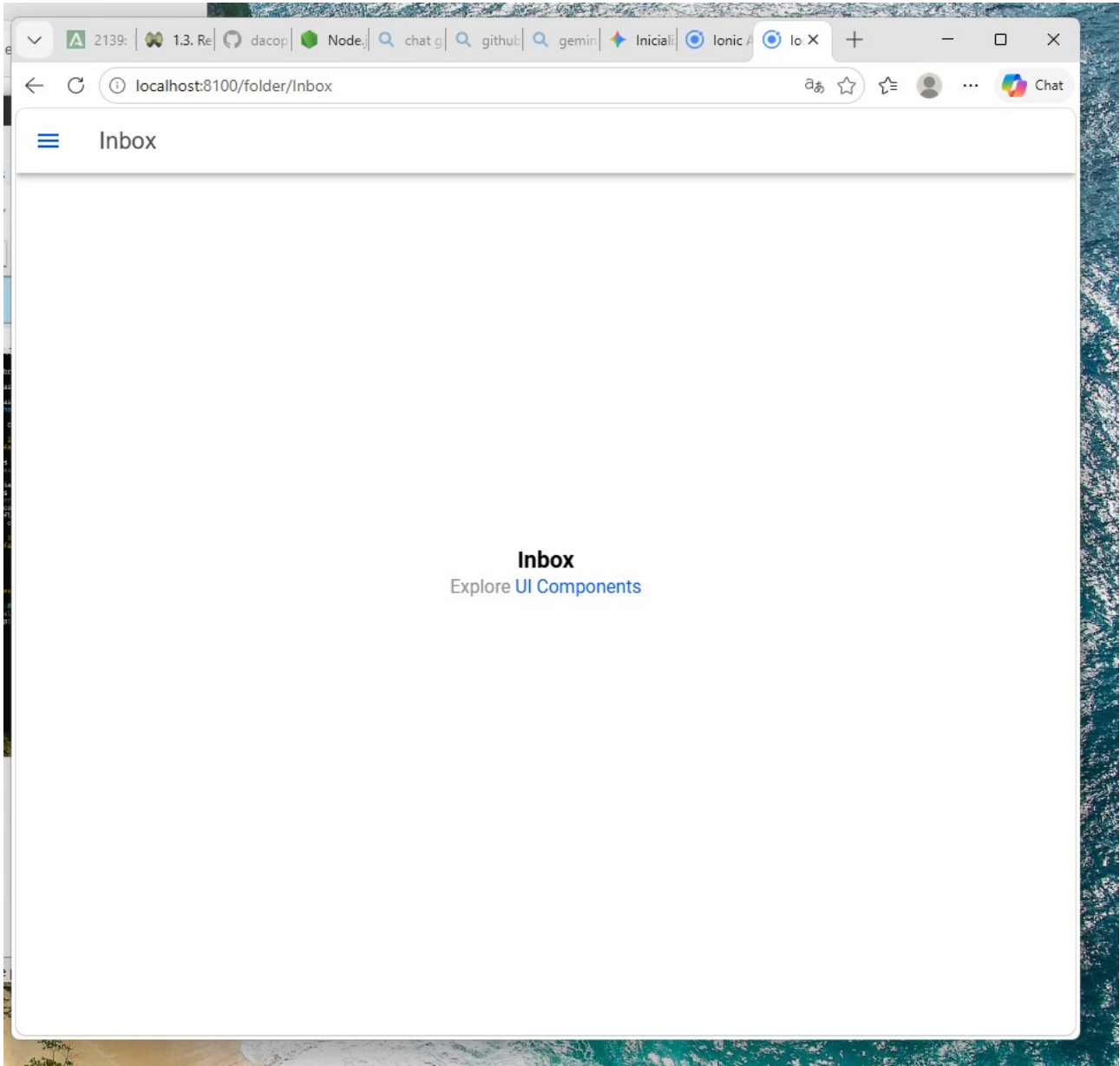
Captura 15



```
C:\Users\06_2DAW> cd aplicacioneshibridas
C:\Users\06_2DAW\aplicacioneshibridas> ./misegundaapp
C:\Users\06_2DAW\aplicacioneshibridas\misegundaapp>ionic serve
> ng.cmd run app:serve --host=localhost --port=8100
[ng] > Building...
[ng] One or more browsers which are configured in the project's Browserslist configuration fall outside Angular's browser support for this version.
[ng] Unsupported browsers:
[ng] chrome 110, chrome 109, chrome 108, chrome 107, edge 110, edge 109, edge 108, edge 107, firefox 111, firefox 110, firefox 109, firefox 108, firefox 107, firefox 106, ios_saf 16.3, ios_saf 16.2, ios_saf 16.1, safari 16.3, safari 16.2, safari 16.1
[ng] / Building...
[ng] Initial chunk files      Names      Raw size
[ng] styles.css               styles    57.70 kB
[ng] main.js                  main      13.54 kB
[ng]                         Initial total 71.24 kB
[ng] Lazy chunk files        Names      Raw size
[ng] folder-page=24581704.js  folder-page 5.47 kB
[ng] Application bundle generation complete. [3.727 seconds] - 2026-09-18T07:53:02.431Z
[ng] NOTE: Raw file sizes do not reflect development server per-request transformations.
[ng] One or more browsers which are configured in the project's Browserslist configuration fall outside Angular's browser support for this version.
[ng] Unsupported browsers:
[ng] chrome 110, chrome 109, chrome 108, chrome 107, edge 110, edge 109, edge 108, edge 107, firefox 111, firefox 110, firefox 109, firefox 108, firefox 107, firefox 106, ios_saf 16.3, ios_saf 16.2, ios_saf 16.1, safari 16.3, safari 16.2, safari 16.1
[INFO] Development server running!
      Local: http://localhost:8100
      Use Ctrl+C to quit this process
[ng] → Local: http://localhost:8100/
[ng] → press h + enter to show help
[INFO] Browser window opened to http://localhost:8100!
```

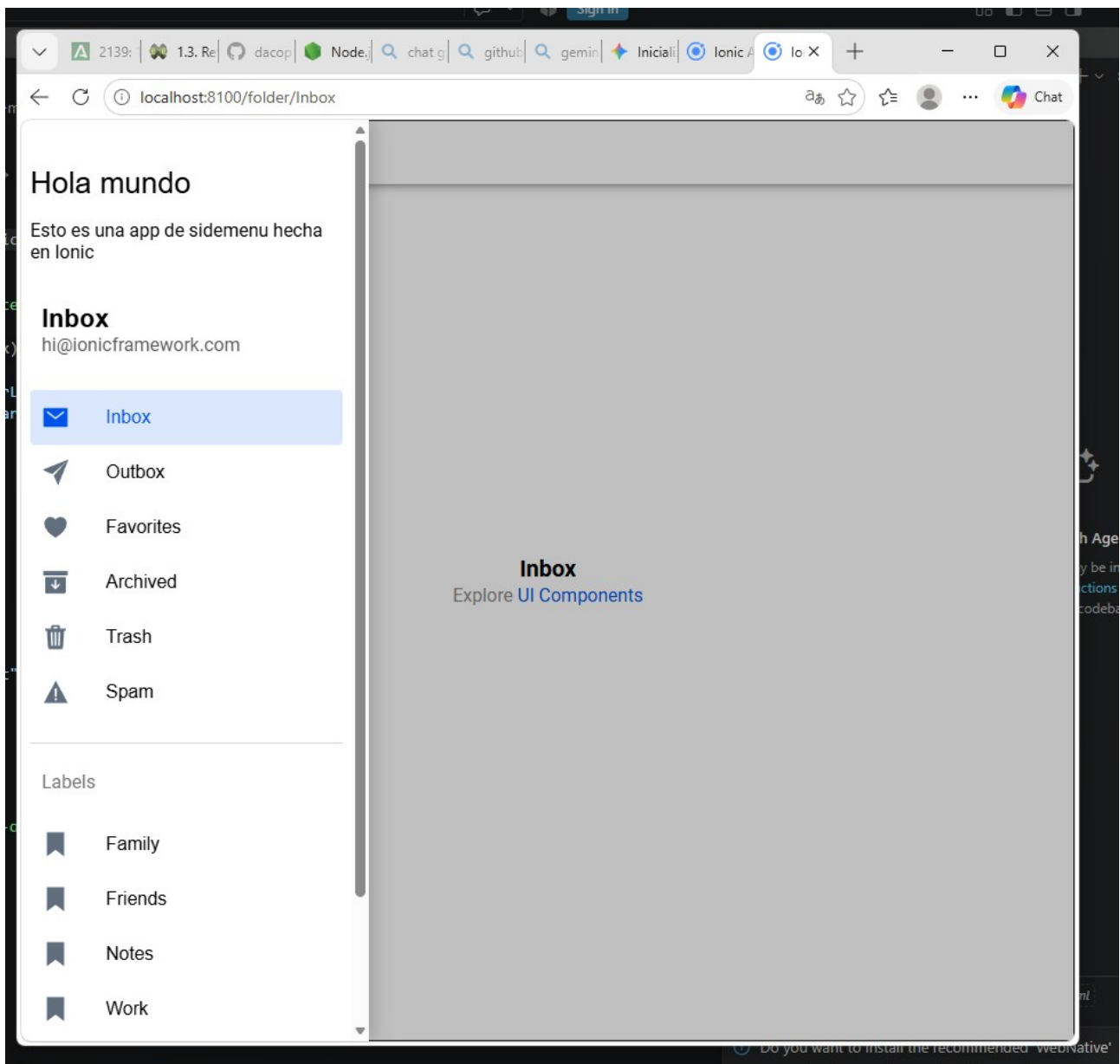
Se entra en la carpeta misegundaapp y se ejecuta ionic serve; la app se compila y el servidor arranca en <http://localhost:8100>.

Captura 16



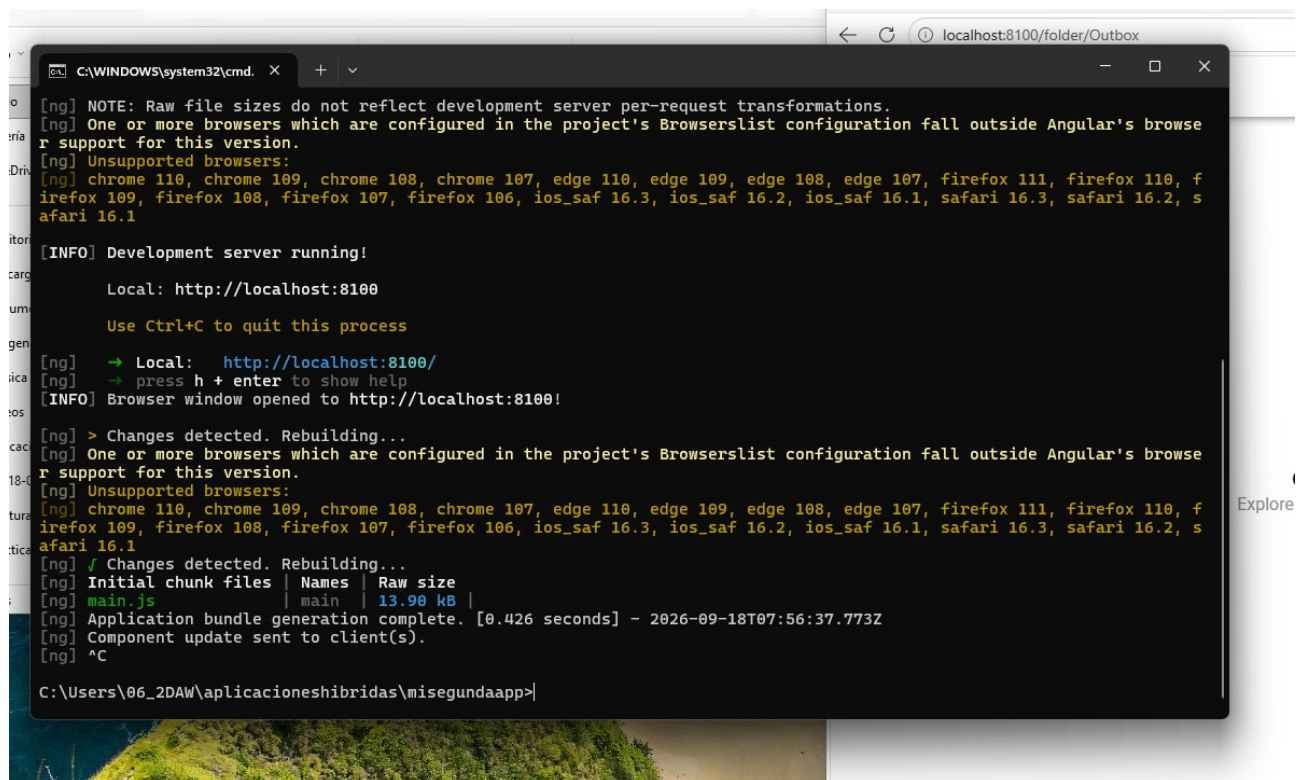
La segunda app (misegundaapp, plantilla sidemenu) corriendo en el navegador, mostrando la página "Inbox" y el icono de menú .

Captura 17



Al pulsar el icono del menú se despliega el menú lateral con las carpetas (Inbox, Outbox, Favorites, Archived, Trash, Spam) y las etiquetas (Family, Friends, Notes, Work).

Captura 18

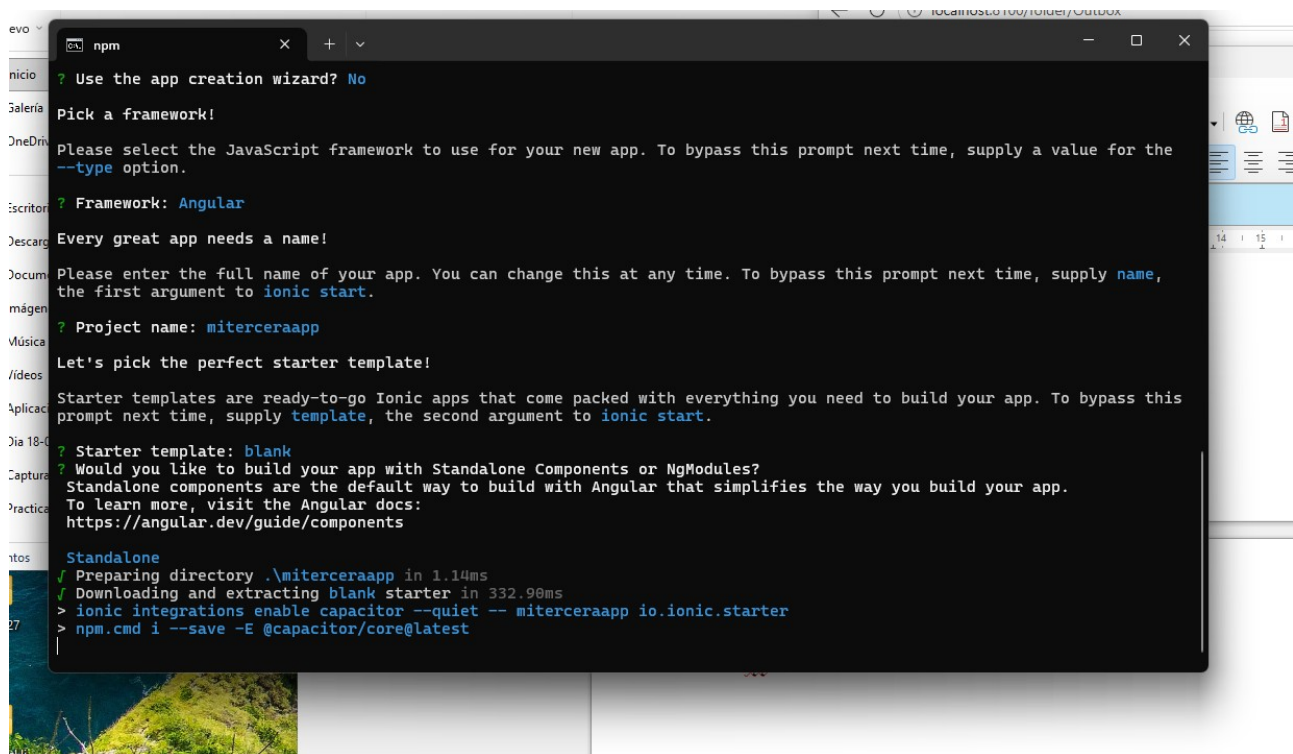


```
C:\WINDOWS\system32\cmd. X + v
[ng] NOTE: Raw file sizes do not reflect development server per-request transformations.
[ng] One or more browsers which are configured in the project's Browserslist configuration fall outside Angular's browser support for this version.
[ng] Unsupported browsers:
[ng] chrome 110, chrome 109, chrome 108, chrome 107, edge 110, edge 109, edge 108, edge 107, firefox 111, firefox 110, firefox 109, firefox 108, firefox 107, firefox 106, ios_saf 16.3, ios_saf 16.2, ios_saf 16.1, safari 16.3, safari 16.2, safari 16.1
[INFO] Development server running!
Local: http://localhost:8100
Use Ctrl+C to quit this process
[ng] → Local: http://localhost:8100/
[ng] → press h + enter to show help
[INFO] Browser window opened to http://localhost:8100!
[ng] > Changes detected. Rebuilding...
[ng] One or more browsers which are configured in the project's Browserslist configuration fall outside Angular's browser support for this version.
[ng] Unsupported browsers:
[ng] chrome 110, chrome 109, chrome 108, chrome 107, edge 110, edge 109, edge 108, edge 107, firefox 111, firefox 110, firefox 109, firefox 108, firefox 107, firefox 106, ios_saf 16.3, ios_saf 16.2, ios_saf 16.1, safari 16.3, safari 16.2, safari 16.1
[ng] ✓ Changes detected. Rebuilding...
[ng] Initial chunk files | Names | Raw size
[ng] main.js | main | 13.90 kB |
[ng] Application bundle generation complete. [0.426 seconds] - 2026-09-18T07:56:37.773Z
[ng] Component update sent to client(s).
[ng] ^C
C:\Users\06_2DAW\aplicacioneshibridas\misegundaapp>
```

En la terminal se ve el servidor deteniéndose con Ctrl+C tras detectar y recompilar varios cambios.

Mi tercera app

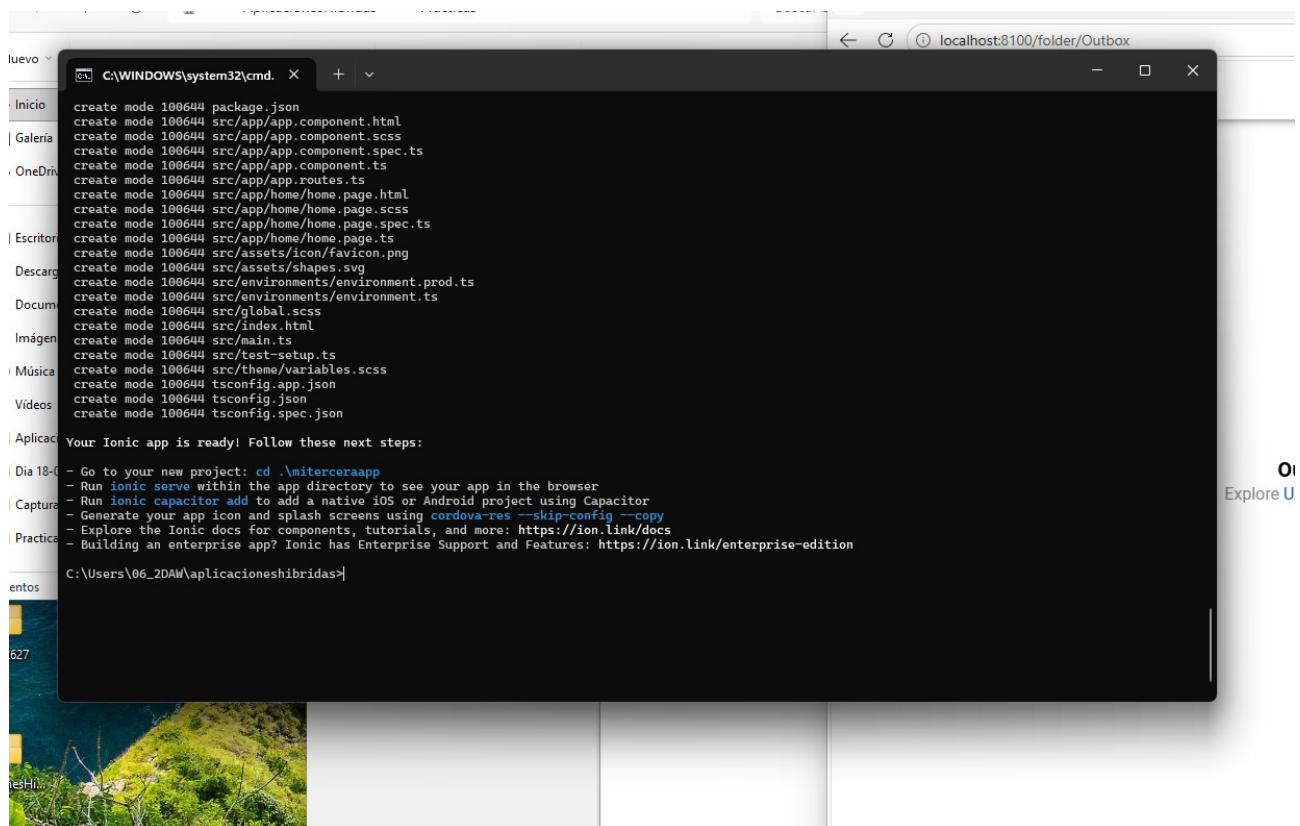
Captura 19



```
npm
? Use the app creation wizard? No
Pick a framework!
Please select the JavaScript framework to use for your new app. To bypass this prompt next time, supply a value for the
--type option.
? Framework: Angular
Every great app needs a name!
Please enter the full name of your app. You can change this at any time. To bypass this prompt next time, supply name,
the first argument to ionic start.
? Project name: miterceraapp
Let's pick the perfect starter template!
Starter templates are ready-to-go Ionic apps that come packed with everything you need to build your app. To bypass this
prompt next time, supply template, the second argument to ionic start.
? Starter template: blank
? Would you like to build your app with Standalone Components or NgModules?
Standalone components are the default way to build with Angular that simplifies the way you build your app.
To learn more, visit the Angular docs:
https://angular.dev/guide/components
Standalone
✓ Preparing directory .\miterceraapp in 1.14ms
✓ Downloading and extracting blank starter in 332.90ms
> ionic integrations enable capacitor --quiet -- miterceraapp io.ionic.starter
> npm.cmd i --save -E @capacitor/core@latest
```

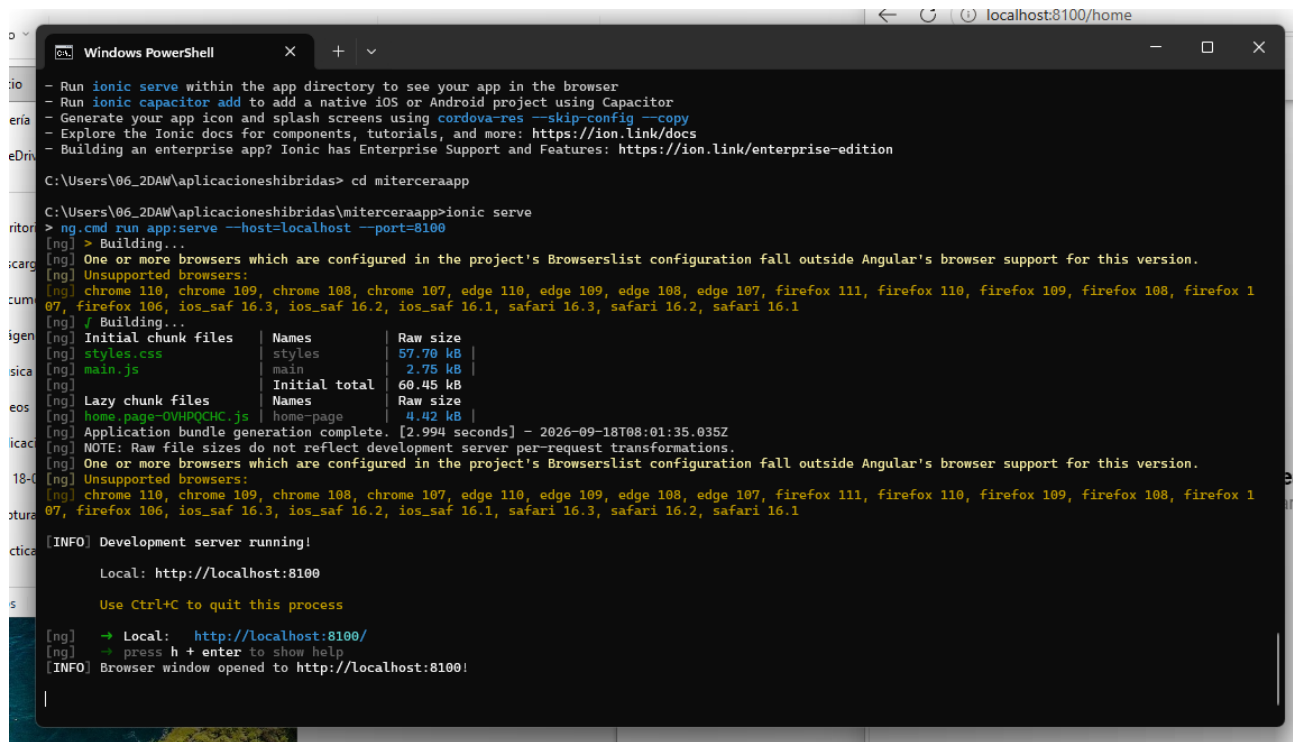
Inicio de la tercera app mediante el asistente interactivo de ionic start: framework Angular, nombre "miterceraapp", plantilla "blank" y Standalone Components.

Captura 20



Finaliza la creación de "miterceraapp", mostrando los archivos generados y los pasos siguientes sugeridos por Ionic.

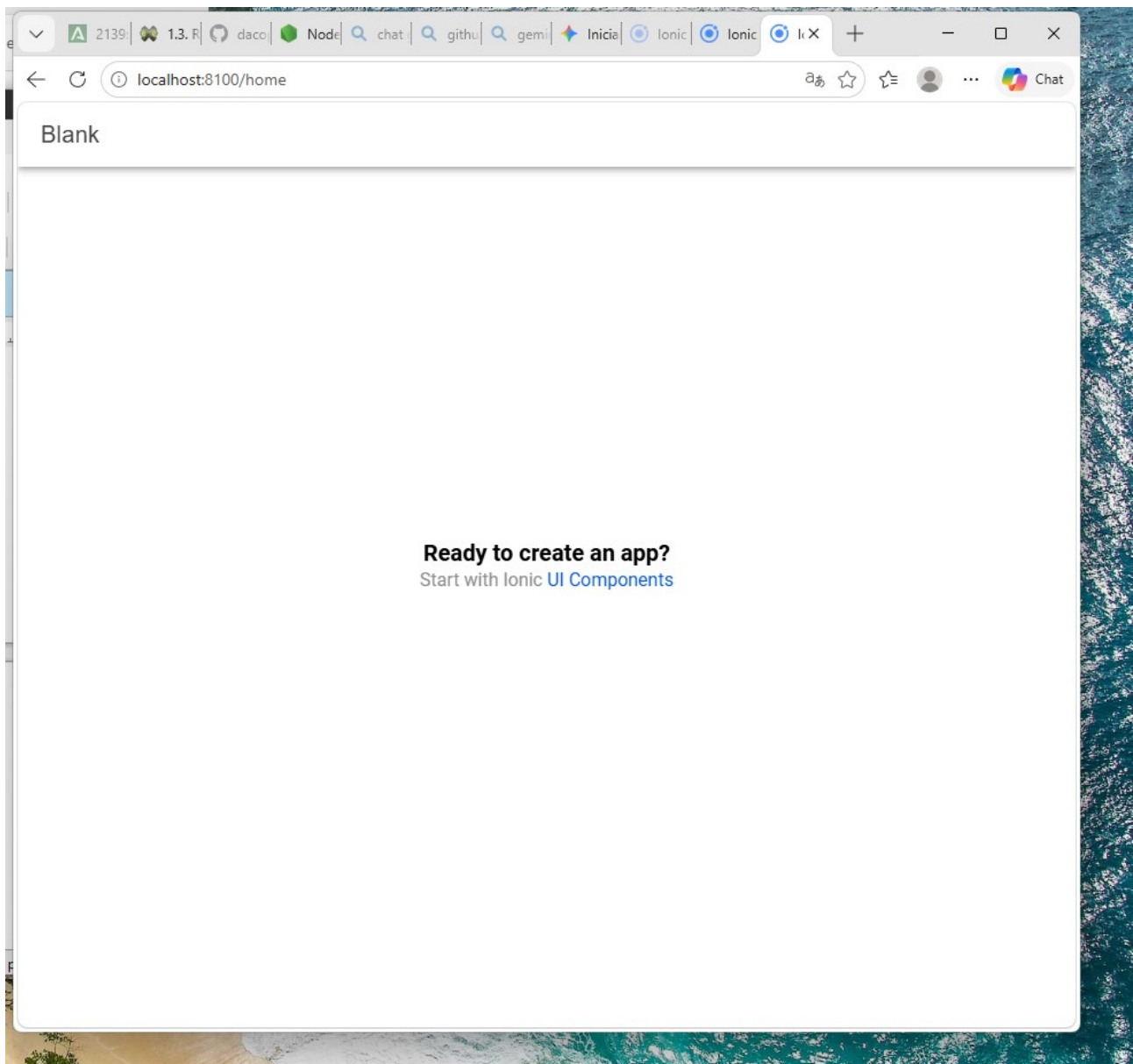
Captura 21



```
C:\Users\06_2DAW\aplicacioneshibridas> cd miterceraapp
C:\Users\06_2DAW\aplicacioneshibridas\miterceraapp> ionic serve
> ng.cmd run app:serve --host=localhost --port=8100
[ng] > Building...
[ng] One or more browsers which are configured in the project's Browserslist configuration fall outside Angular's browser support for this version.
[ng] Unsupported browsers:
[ng] chrome 110, chrome 109, chrome 108, chrome 107, edge 110, edge 109, edge 108, edge 107, firefox 111, firefox 110, firefox 109, firefox 108, firefox 107,
[ng] firefox 106, ios_saf 16.3, ios_saf 16.2, ios_saf 16.1, safari 16.3, safari 16.2, safari 16.1
[ng] / Building...
[ng] Initial chunk files | Names | Raw size
[ng] styles.css | styles | 57.70 kB
[ng] main.js | main | 2.75 kB
[ng] | Initial total | 60.45 kB
[ng] Lazy chunk files | Names | Raw size
[ng] home.page-0VH8QCHC.js | home-page | 4.42 kB
[ng] Application bundle generation complete. [2.994 seconds] - 2026-09-18T08:01:35.035Z
[ng] NOTE: Raw file sizes do not reflect development server per-request transformations.
[ng] One or more browsers which are configured in the project's Browserslist configuration fall outside Angular's browser support for this version.
[ng] Unsupported browsers:
[ng] chrome 110, chrome 109, chrome 108, chrome 107, edge 110, edge 109, edge 108, edge 107, firefox 111, firefox 110, firefox 109, firefox 108, firefox 107,
[ng] firefox 106, ios_saf 16.3, ios_saf 16.2, ios_saf 16.1, safari 16.3, safari 16.2, safari 16.1
[INFO] Development server running!
Local: http://localhost:8100
Use Ctrl+C to quit this process
[ng] → Local: http://localhost:8100/
[ng] → press h + enter to show help
[INFO] Browser window opened to http://localhost:8100!
```

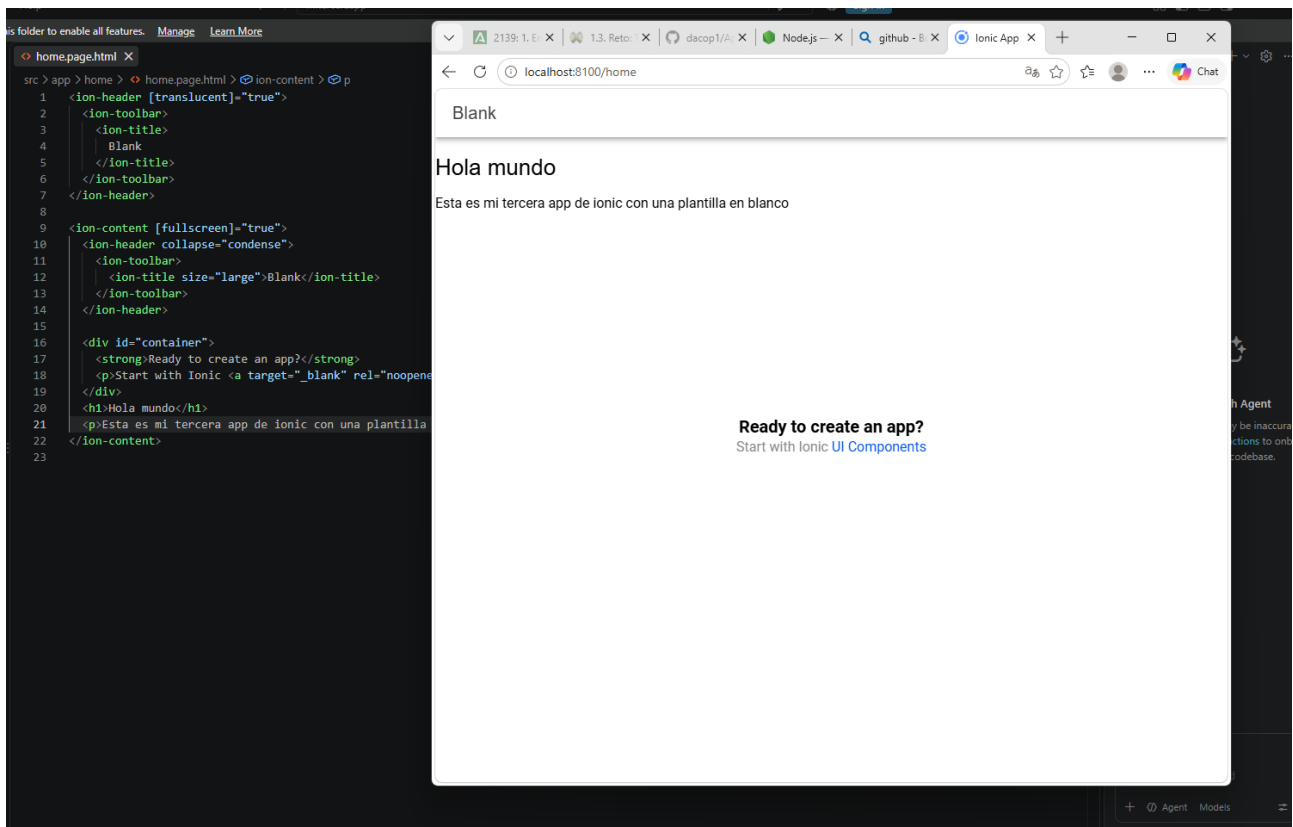
Se accede a la carpeta miterceraapp y se ejecuta ionic serve; se compila el bundle y arranca el servidor en <http://localhost:8100>.

Captura 22



La tercera app abierta en el navegador (localhost:8100/home), mostrando la plantilla "Blank" por defecto.

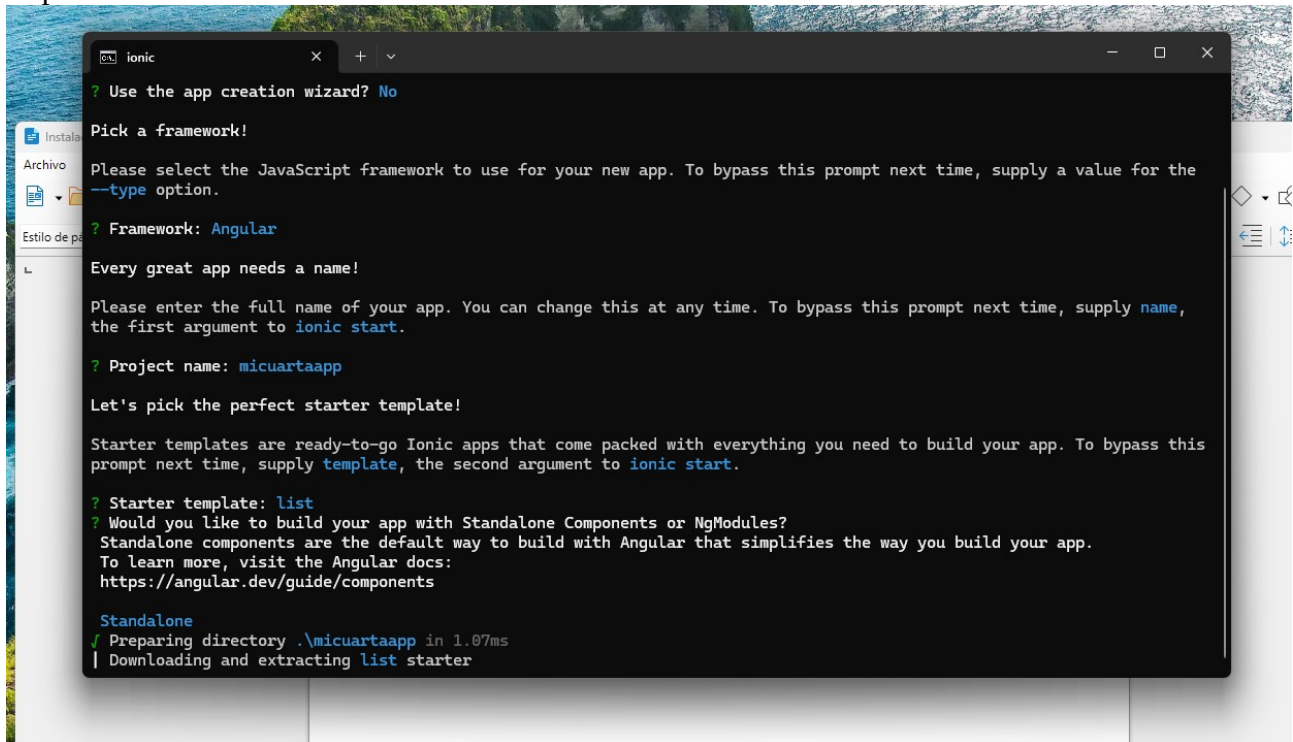
Captura 23



Se edita `home.page.html` añadiendo el texto "Hola mundo, esta es mi tercera app de Ionic con una plantilla en blanco", visible en el navegador.

mi cuarta app

Captura24



```
ionic
? Use the app creation wizard? No

Pick a framework!
Please select the JavaScript framework to use for your new app. To bypass this prompt next time, supply a value for the
--type option.
? Framework: Angular

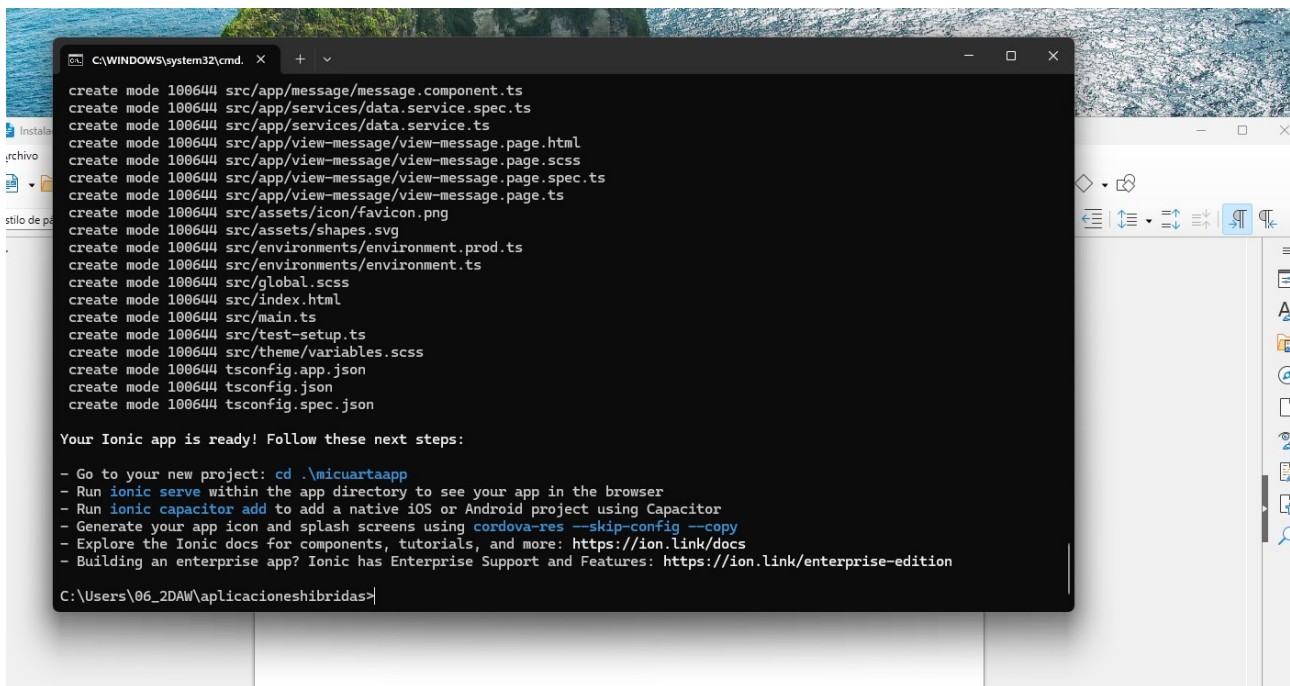
Every great app needs a name!
Please enter the full name of your app. You can change this at any time. To bypass this prompt next time, supply name,
the first argument to ionic start.
? Project name: micuartaapp

Let's pick the perfect starter template!
Starter templates are ready-to-go Ionic apps that come packed with everything you need to build your app. To bypass this
prompt next time, supply template, the second argument to ionic start.
? Starter template: list
? Would you like to build your app with Standalone Components or NgModules?
Standalone components are the default way to build with Angular that simplifies the way you build your app.
To learn more, visit the Angular docs:
https://angular.dev/guide/components

Standalone
✓ Preparing directory .\micuartaapp in 1.07ms
| Downloading and extracting list starter
```

Creación de la cuarta app ("micuartaapp") con el asistente de ionic start, eligiendo la plantilla "list".

Captura 25



```
C:\WINDOWS\system32\cmd. x + v
create mode 100644 src/app/message/message.component.ts
create mode 100644 src/app/services/data.service.spec.ts
create mode 100644 src/app/services/data.service.ts
create mode 100644 src/app/view-message/view-message.page.html
create mode 100644 src/app/view-message/view-message.page.scss
create mode 100644 src/app/view-message/view-message.page.spec.ts
create mode 100644 src/app/view-message/view-message.page.ts
create mode 100644 src/assets/icon/favicon.png
create mode 100644 src/assets/shapes.svg
create mode 100644 src/environments/environment.prod.ts
create mode 100644 src/environments/environment.ts
create mode 100644 src/global.scss
create mode 100644 src/index.html
create mode 100644 src/main.ts
create mode 100644 src/test-setup.ts
create mode 100644 src/theme/variables.scss
create mode 100644 tsconfig.app.json
create mode 100644 tsconfig.json
create mode 100644 tsconfig.spec.json

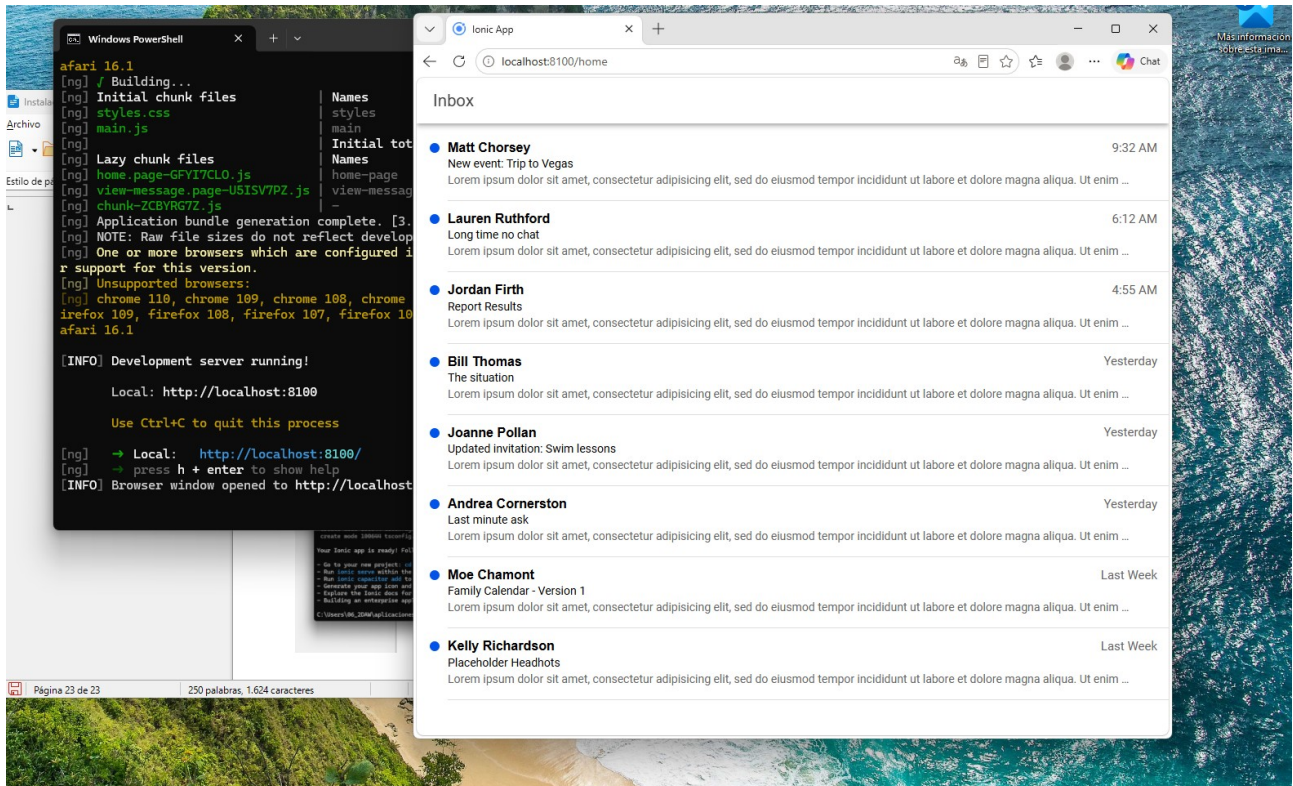
Your Ionic app is ready! Follow these next steps:

- Go to your new project: cd .\micuartaapp
- Run ionic serve within the app directory to see your app in the browser
- Run ionic capacitor add to add a native iOS or Android project using Capacitor
- Generate your app icon and splash screens using cordova-res --skip-config --copy
- Explore the Ionic docs for components, tutorials, and more: https://ion.link/docs
- Building an enterprise app? Ionic has Enterprise Support and Features: https://ion.link/enterprise-edition

C:\Users\06_2DAW\aplicacioneshibridas>
```

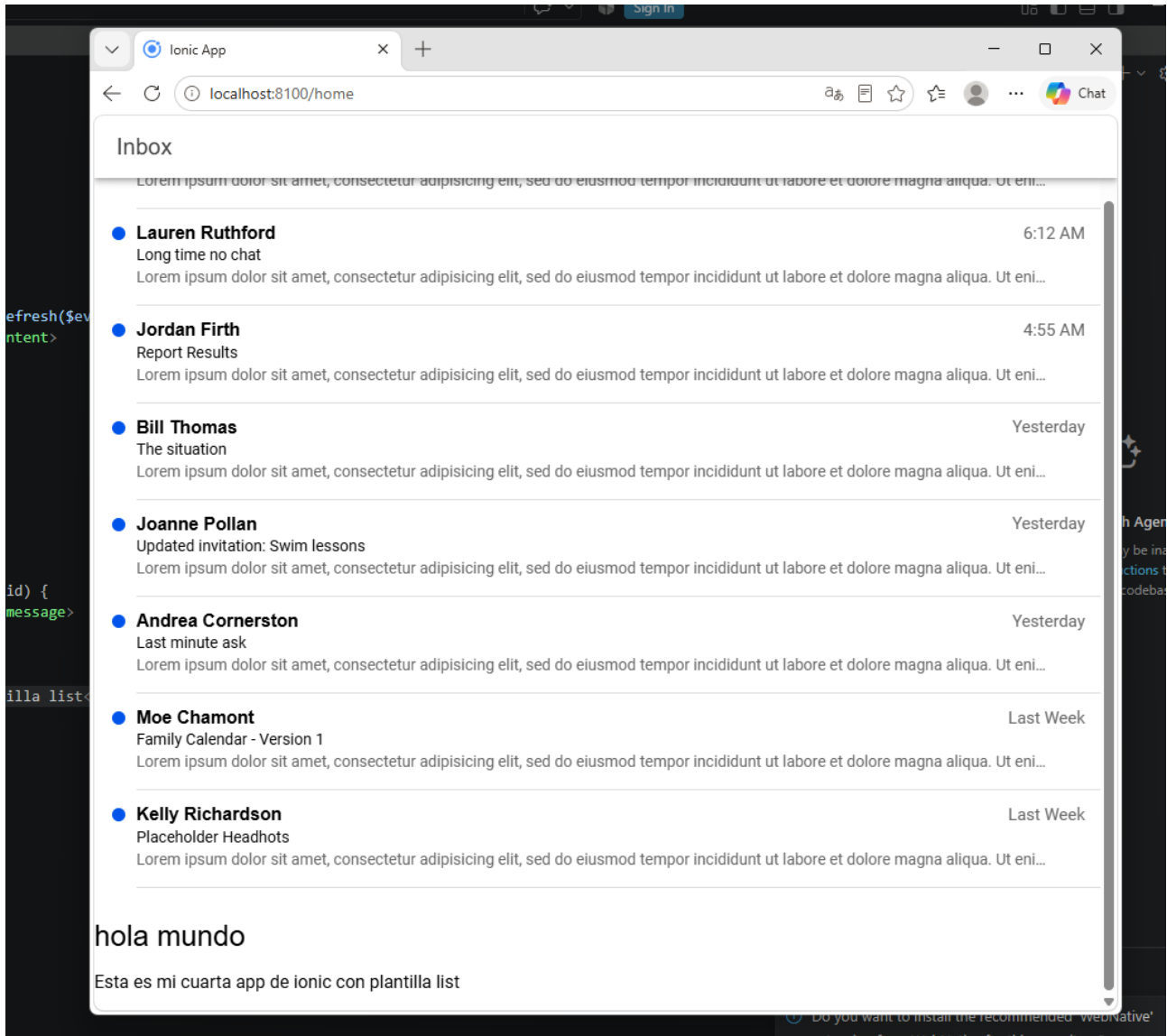
Finaliza la creación de "micuartaapp"; se generan los archivos propios de la plantilla list (message.component, data.service, view-message.page, etc.).

Captura 26



Al ejecutar ionic serve se abre el navegador mostrando la plantilla "list" ya funcional: una bandeja de entrada con una lista de mensajes de ejemplo.

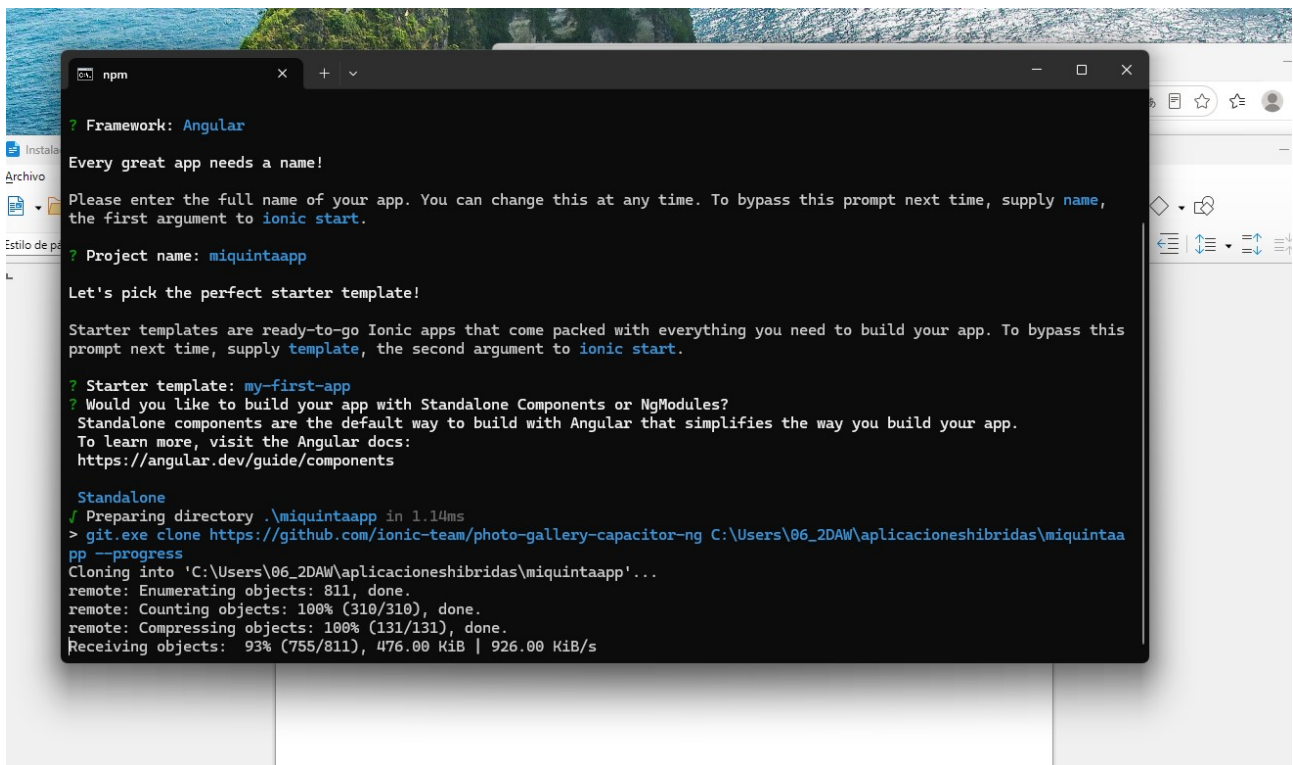
Captura 27



Se añade el texto personalizado "hola mundo, esta es mi cuarta app de ionic con plantilla list" al final de la lista de mensajes.

mi quinta app

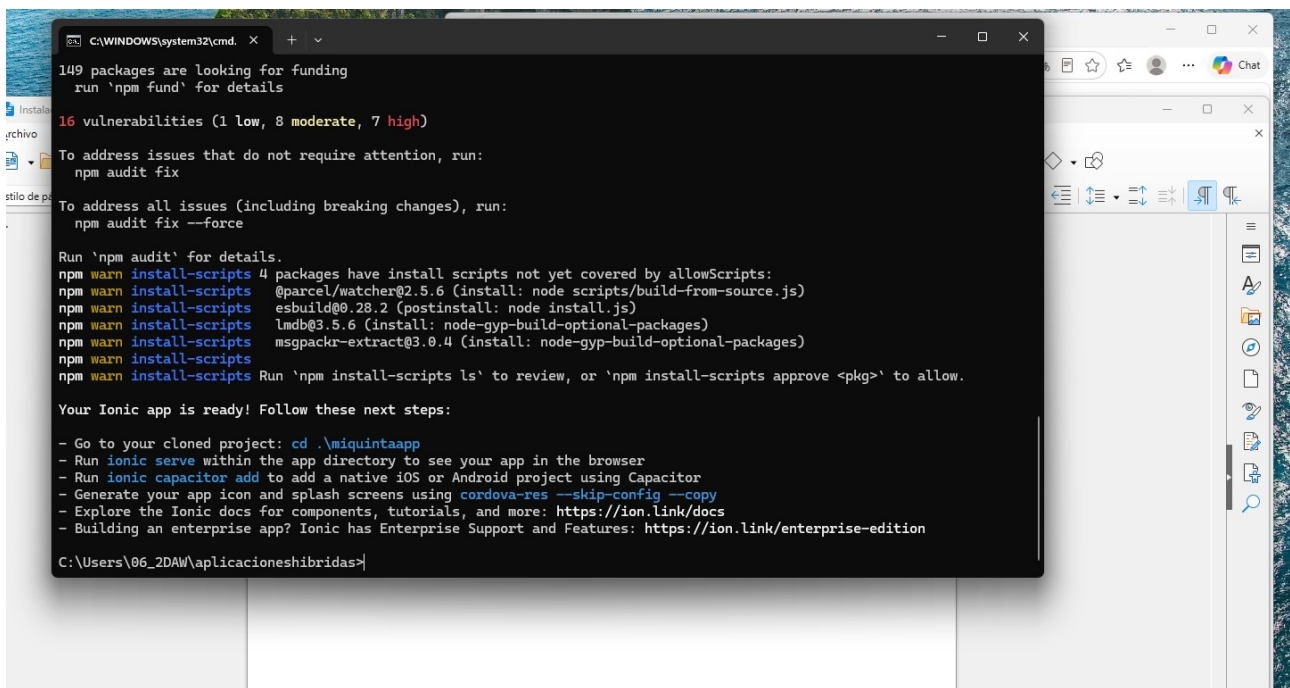
Captura 28



```
npm
? Framework: Angular
Every great app needs a name!
Please enter the full name of your app. You can change this at any time. To bypass this prompt next time, supply name,
the first argument to ionic start.
? Project name: miquintaapp
Let's pick the perfect starter template!
Starter templates are ready-to-go Ionic apps that come packed with everything you need to build your app. To bypass this
prompt next time, supply template, the second argument to ionic start.
? Starter template: my-first-app
? Would you like to build your app with Standalone Components or NgModules?
Standalone components are the default way to build with Angular that simplifies the way you build your app.
To learn more, visit the Angular docs:
https://angular.dev/guide/components
Standalone
✓ Preparing directory .\miquintaapp in 1.14ms
> git.exe clone https://github.com/ionic-team/photo-gallery-capacitor-ng C:\Users\06_2DAW\aplicacioneshibridas\miquintaapp --progress
Cloning into 'C:\Users\06_2DAW\aplicacioneshibridas\miquintaapp'...
remote: Enumerating objects: 811, done.
remote: Counting objects: 100% (310/310), done.
remote: Compressing objects: 100% (131/131), done.
Receiving objects: 93% (755/811), 476.00 KiB | 926.00 KiB/s
```

Creación de la quinta app ("miquintaapp") con la plantilla "my-first-app", que en realidad clona un repositorio de ejemplo de GitHub (photo-gallery-capacitor-ng).

Captura 29



```
C:\WINDOWS\system32\cmd.exe
149 packages are looking for funding
  run 'npm fund' for details

16 vulnerabilities (1 low, 8 moderate, 7 high)

To address issues that do not require attention, run:
  npm audit fix

To address all issues (including breaking changes), run:
  npm audit fix --force

Run 'npm audit' for details.
npm warn install-scripts 4 packages have install scripts not yet covered by allowScripts:
npm warn install-scripts   @parcel/watcher@2.5.6 (install: node scripts/build-from-source.js)
npm warn install-scripts   esbuild@0.28.2 (postinstall: node install.js)
npm warn install-scripts   lmdb@3.5.6 (install: node-gyp-build-optional-packages)
npm warn install-scripts   msgpackr-extract@3.0.4 (install: node-gyp-build-optional-packages)
npm warn install-scripts Run 'npm install-scripts ls' to review, or 'npm install-scripts approve <pkg>' to allow.

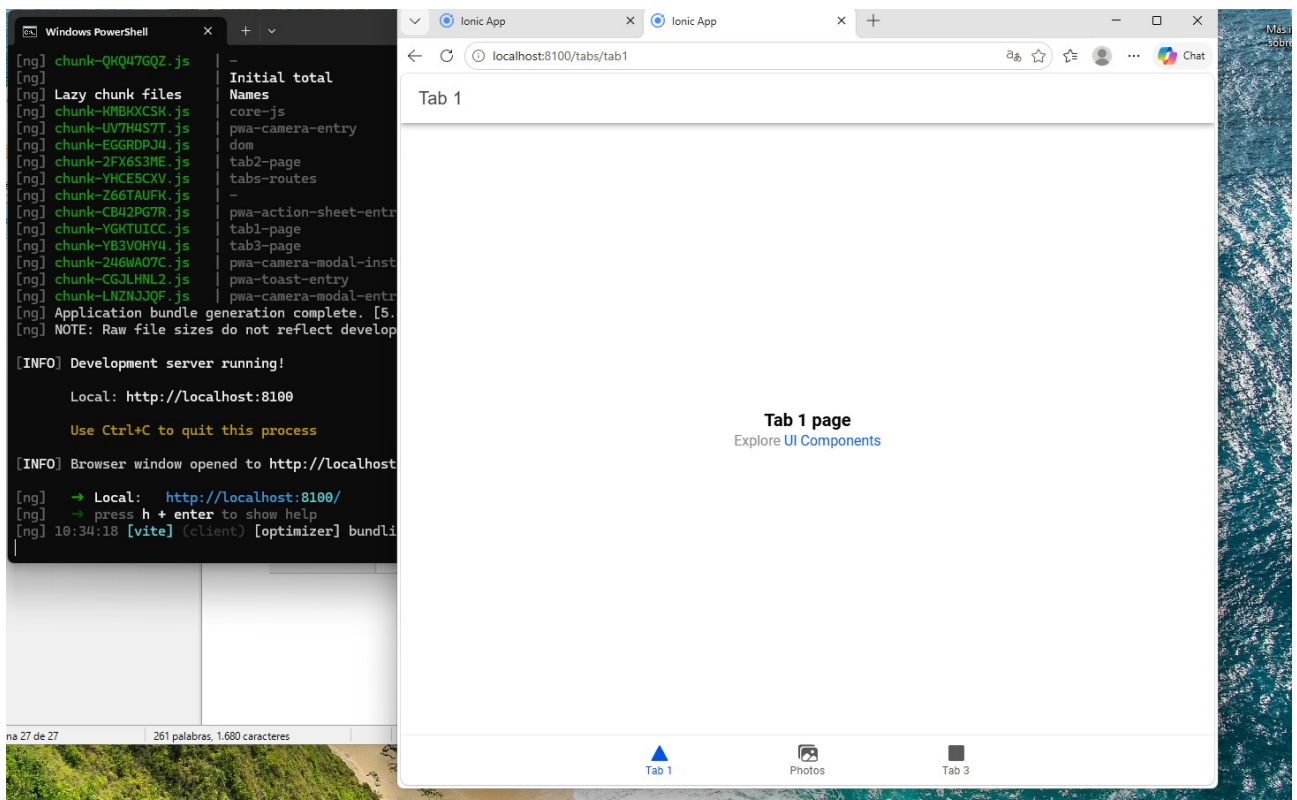
Your Ionic app is ready! Follow these next steps:

- Go to your cloned project: cd .\miquintaapp
- Run ionic serve within the app directory to see your app in the browser
- Run ionic capacitor add to add a native iOS or Android project using Capacitor
- Generate your app icon and splash screens using cordova-res --skip-config --copy
- Explore the Ionic docs for components, tutorials, and more: https://ion.link/docs
- Building an enterprise app? Ionic has Enterprise Support and Features: https://ion.link/enterprise-edition

C:\Users\06_2DAW\aplicacioneshibridas>
```

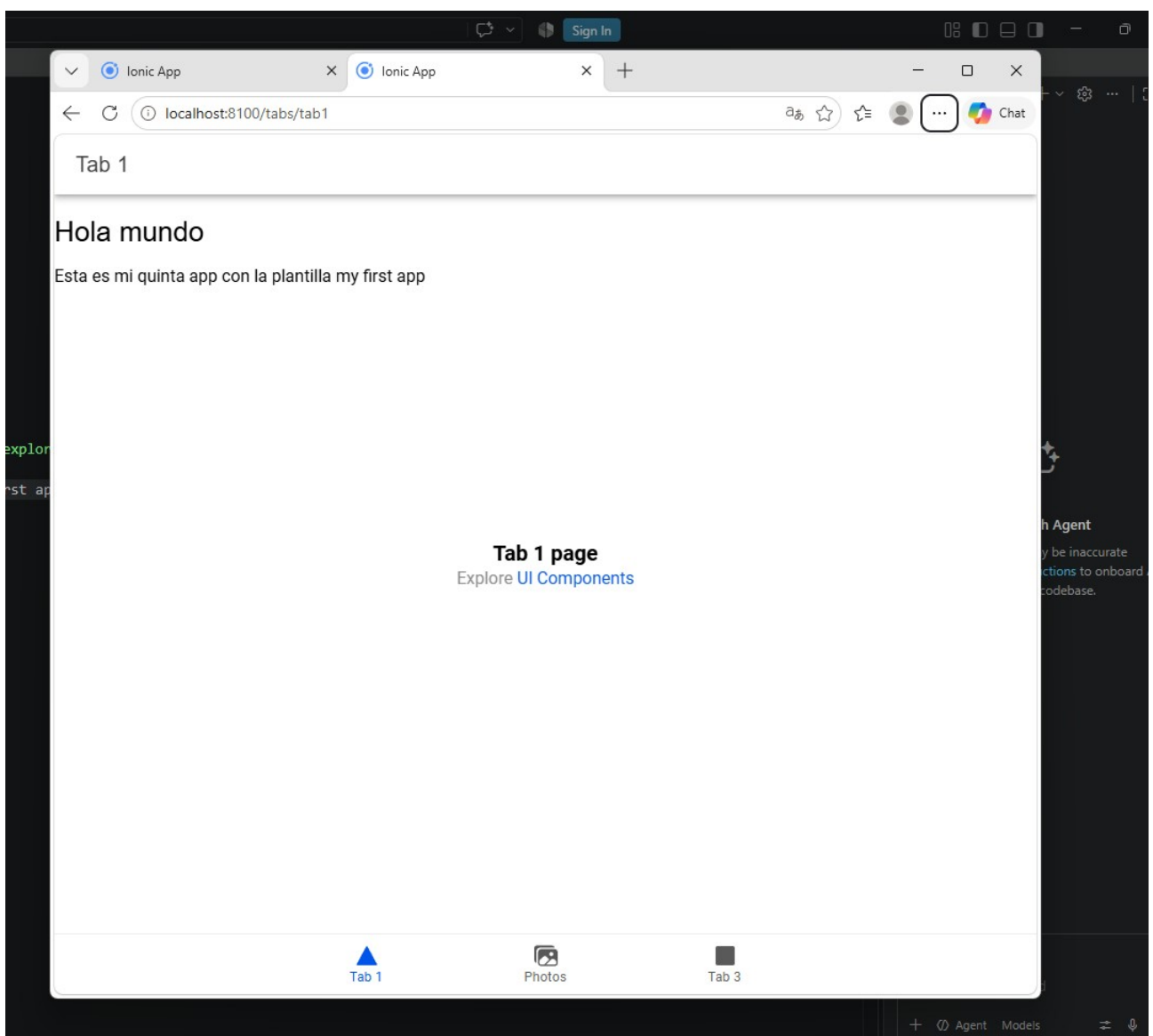
Termina la instalación de dependencias (`npm install`) del proyecto clonado, con avisos habituales de vulnerabilidades, confirmando que el proyecto está listo.

Captura 30



Se ejecuta ionic serve en el proyecto clonado; el bundle incluye módulos de cámara (pwa-camera-entry, pwa-camera-modal, pwa-toast) y la app se abre con las pestañas Tab 1, Photos y Tab 3.

Captura 31

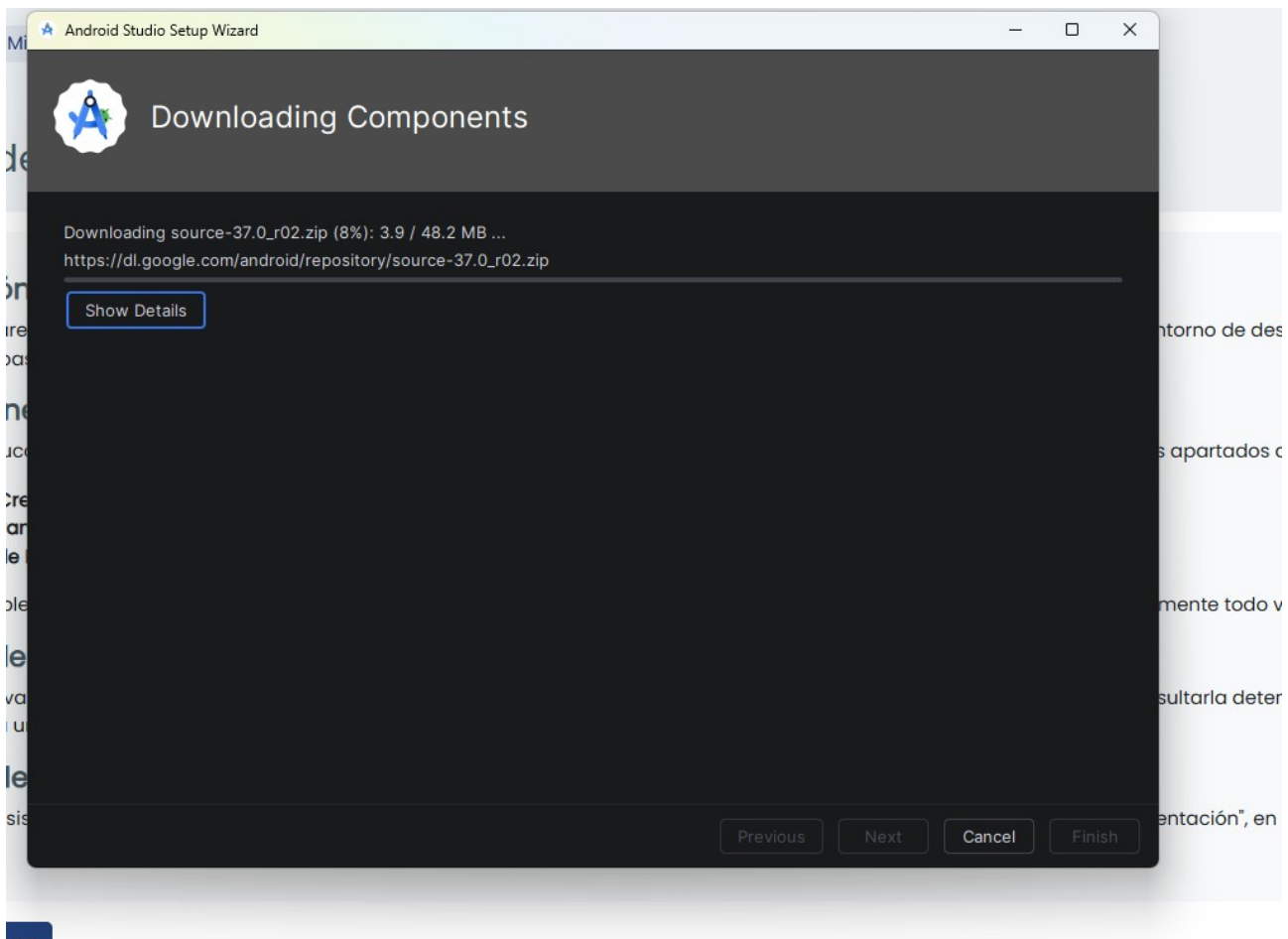


Resultado final tras editar el código: aparece el texto "Hola mundo, esta es mi quinta app con la plantilla my first app".

Día 23/09/2026

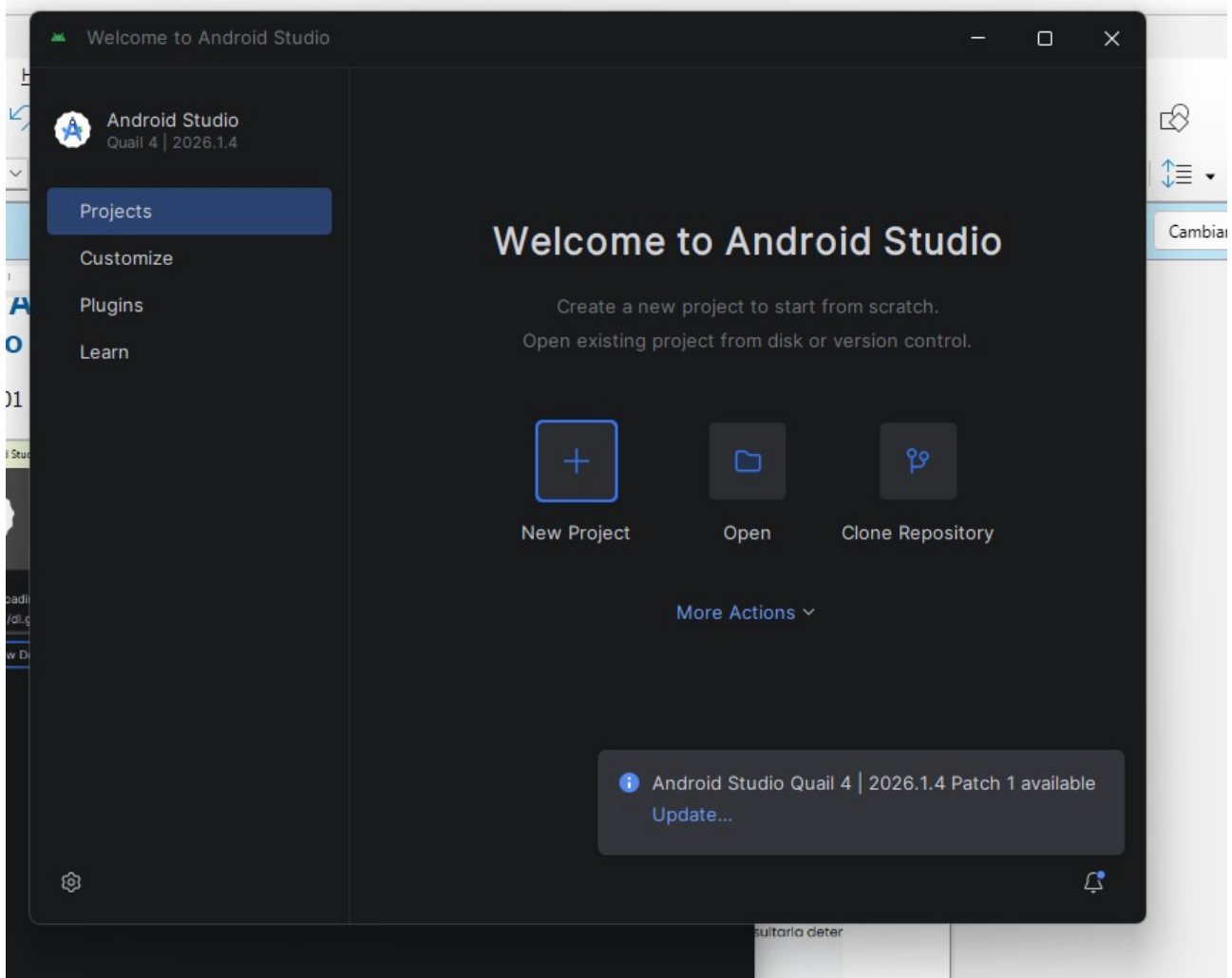
Reto Avanzado: Ejecución en Emulador y Dispositivo Físico

Captura01



Se abre el asistente de instalación de Android Studio, que empieza a descargar los componentes necesarios (SDK, herramientas, etc.).

Captura02



Finaliza la instalación: aparece la pantalla de bienvenida de Android Studio (Quail 4 | 2026.1.4) con las opciones para crear, abrir o clonar un proyecto.

Captura03

Requisitos previos

Capacitor (la herramienta que usa Ionic) requiere una versión específica del JDK para construir aplicaciones de Android. A día de hoy, la recomendada es la **versión 21**.

- Puedes descargar OpenJDK 21 desde la [página oficial de Java](#).
- Descarga el archivo ZIP o el instalador para tu sistema operativo y sigue las instrucciones.
- Asegúrate de que el JDK esté correctamente instalado ejecutando `java -version` en tu terminal. Deberías ver una salida que mencione la versión 21.

Configurar la variable de entorno ANDROID_HOME

Se necesita decirle a tu sistema operativo dónde está instalado el SDK de Android. Esto se hace creando una variable de entorno llamada `ANDROID_HOME`.

- En Windows:**
 - Abre el menú de inicio y busca "Editar las variables de entorno del sistema".
 - En la ventana de "Propiedades del sistema", haz clic en "Variables de entorno...".
 - En la sección "Variables de usuario", haz clic en "Nueva...".
 - Nombre de la variable:** `ANDROID_HOME`
 - Valor de la variable:** La ruta a tu SDK de Android. Por defecto suele ser: `C:\Users\TU_USUARIO\AppData\Local\Android\Sdk` (recuerda cambiar "TU_USUARIO").
 - Acepta todas las ventanas. Es posible que necesites reiniciar la terminal (o el ordenador) para que los cambios surtan efecto.
- En macOS o Linux:**
 - Abre una terminal.
 - Edita el archivo de configuración de tu shell: `~/.bash_profile` o `~/.zshrc`.
 - Añade la siguiente línea al final del archivo: `export ANDROID_HOME=$HOME/Library/Android/sdk`
 - Guarda el archivo (Ctrl+O en nano).
 - Recarga la configuración de tu terminal ejecutando `source ~/.zshrc` o simplemente abriendo una nueva ventana de terminal.

Parte 2: Crear un Emulador de Android

Un emulador es un dispositivo virtual que se ejecuta en tu ordenador. Es perfecto para hacer pruebas rápidas.

- En la pantalla de bienvenida de Android Studio, haz clic en "More Actions" -> "Virtual Device Manager".
- Haz clic en "Create device".
- Elige un modelo de teléfono (por ejemplo, "Pixel 7") y haz clic en "Next".
- Se te mostrará una lista de imágenes del sistema Android. Elige la última versión estable (sin la etiqueta "Tiramisu" o "Upside Down Cake" si aparecen, busca una con un nombre de API numérico alto) y haz clic en el botón "Use".

Captura04

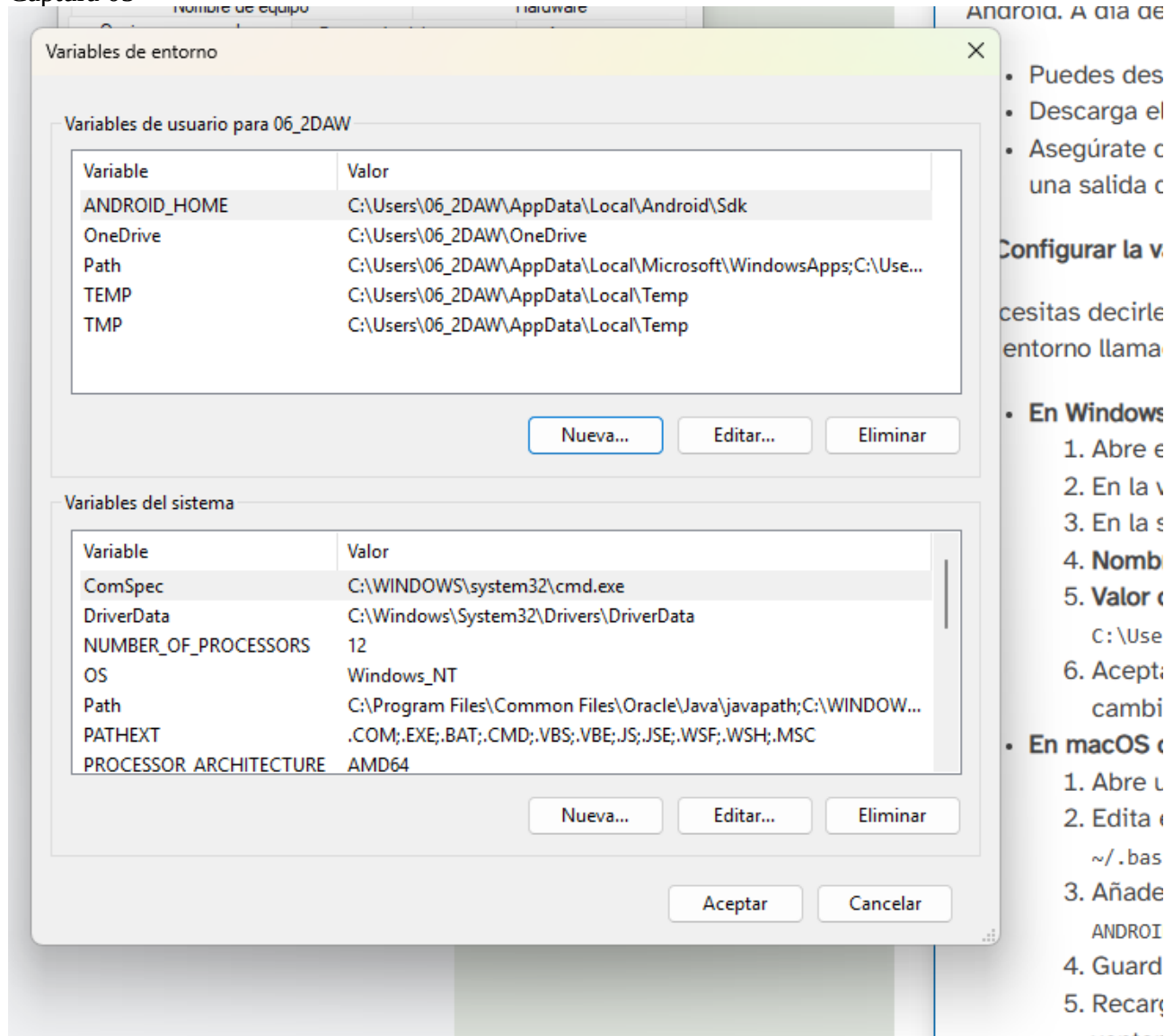
```
Microsoft Windows [Versión 10.0.26100.9457]
(c) Microsoft Corporation. Todos los derechos reservados.

C:\Users\06_2DAW>java --version
java 21.0.7 2025-04-15 LTS
Java(TM) SE Runtime Environment (build 21.0.7+8-LTS-245)
Java HotSpot(TM) 64-Bit Server VM (build 21.0.7+8-LTS-245, mixed mode, sharing)

C:\Users\06_2DAW>
```

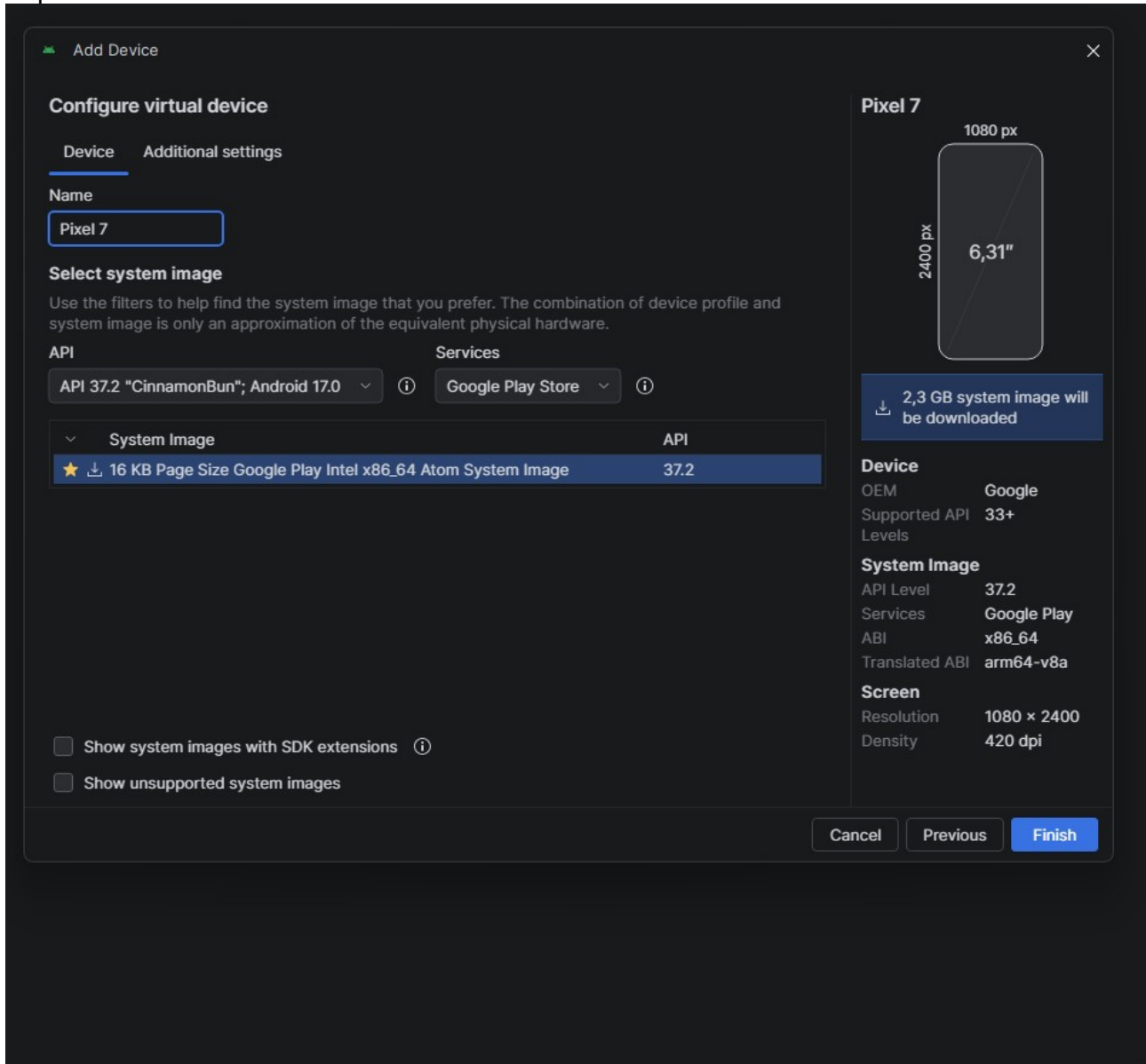
Se verifica en el símbolo del sistema que Java está correctamente instalado (java --version), confirmando la versión 21.0.7 LTS.

Captura 05



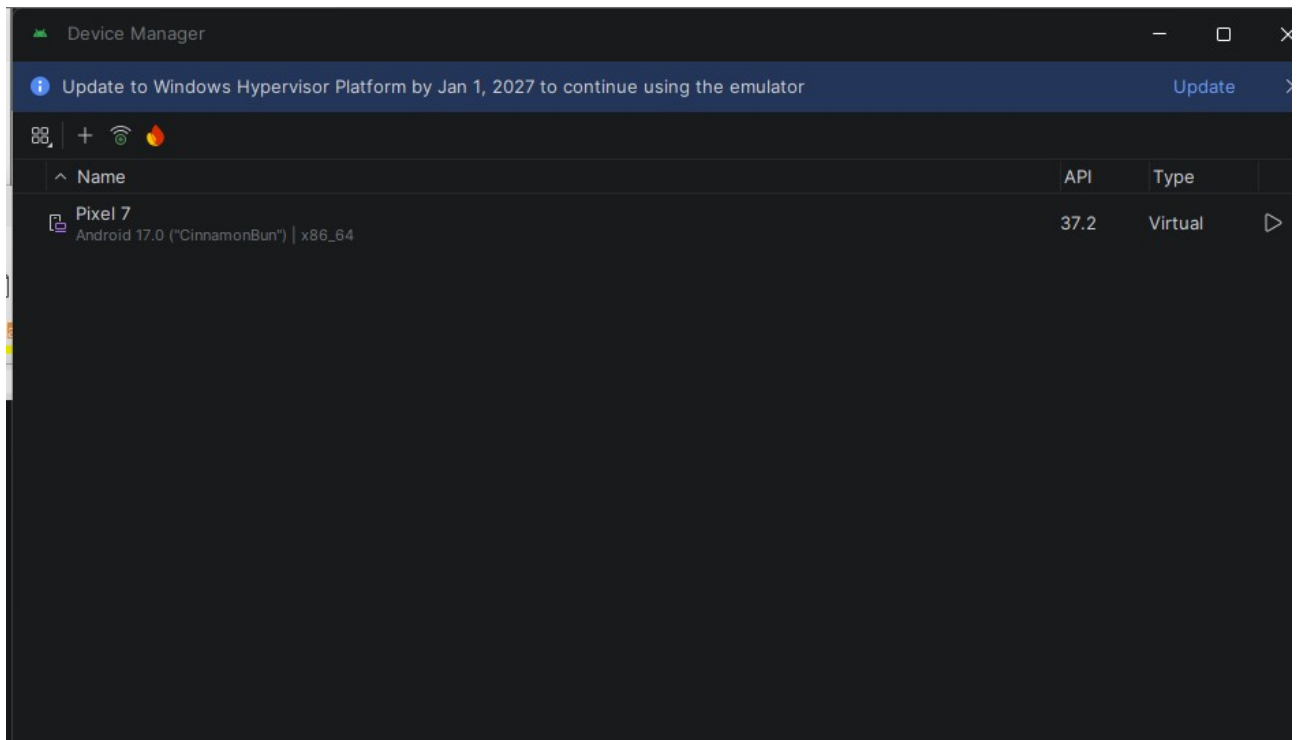
Configuración de las variables de entorno en Windows: se añade la variable ANDROID_HOME apuntando a la carpeta del SDK de Android.

Captura06



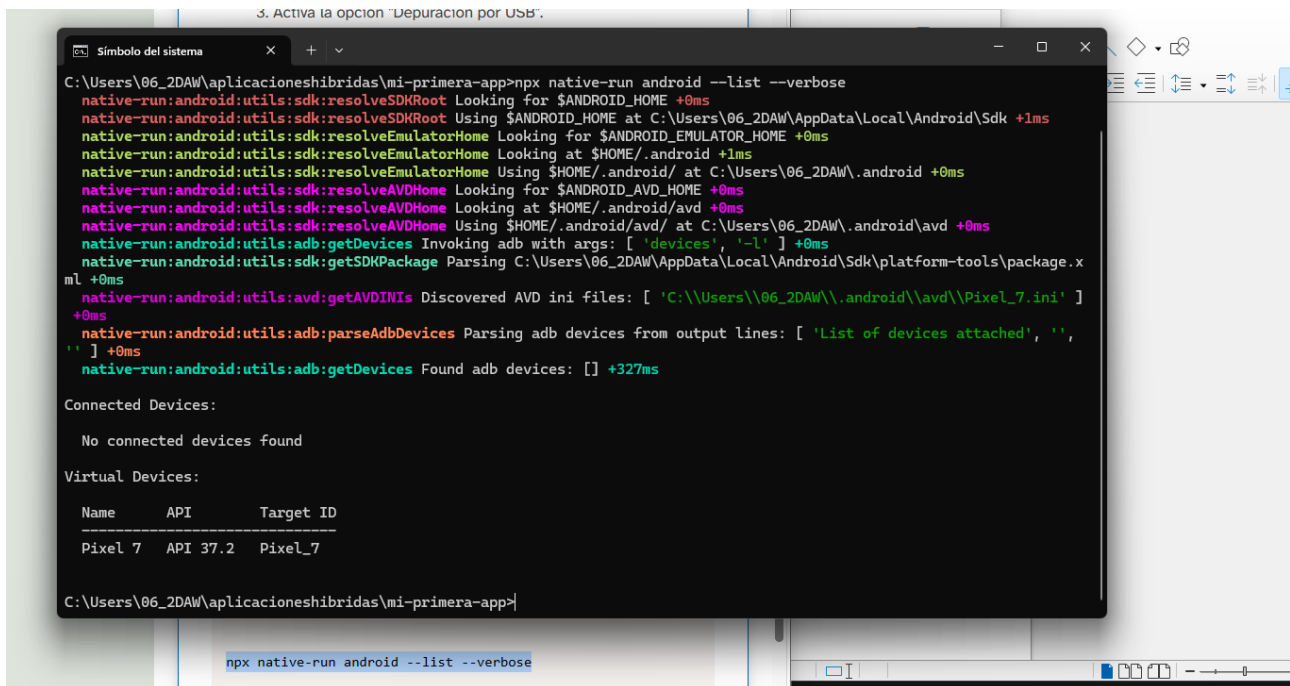
Creación de un dispositivo virtual (AVD) desde el Device Manager: se configura un Pixel 7 con la imagen de sistema API 37.2 (CinnamonBun, Android 17.0) y Google Play Store.

Captura07



El emulador Pixel 7 (Android 17.0, API 37.2) queda creado y disponible en el Device Manager, listo para ejecutarse.

Captura08



```
C:\Users\06_2DAW\aplicacioneshibridas\mi-primer-app>npx native-run android --list --verbose
native-run:android:utils:sdk:resolveSDKRoot Looking for $ANDROID_HOME +0ms
native-run:android:utils:sdk:resolveSDKRoot Using $ANDROID_HOME at C:\Users\06_2DAW\AppData\Local\Android\Sdk +1ms
native-run:android:utils:sdk:resolveEmulatorHome Looking for $ANDROID_EMULATOR_HOME +0ms
native-run:android:utils:sdk:resolveEmulatorHome Looking at $HOME/.android +1ms
native-run:android:utils:sdk:resolveEmulatorHome Using $HOME/.android/ at C:\Users\06_2DAW\.android +0ms
native-run:android:utils:sdk:resolveAVDHome Looking for $ANDROID_AV_HOME +0ms
native-run:android:utils:sdk:resolveAVDHome Looking at $HOME/.android/avd +0ms
native-run:android:utils:sdk:resolveAVDHome Using $HOME/.android/avd/ at C:\Users\06_2DAW\.android\avd +0ms
native-run:android:utils:adb:getDevices Invoking adb with args: [ 'devices', '-l' ] +0ms
native-run:android:utils:sdk:getSDKPackage Parsing C:\Users\06_2DAW\AppData\Local\Android\Sdk\platform-tools\package.x
ml +0ms
native-run:android:utils:avd:getAVDINIIs Discovered AVD ini files: [ 'C:\\Users\\06_2DAW\\.android\\avd\\Pixel_7.ini' ]
+0ms
native-run:android:utils:adb:parseAdbDevices Parsing adb devices from output lines: [ 'List of devices attached', '',
'' ] +0ms
native-run:android:utils:adb:getDevices Found adb devices: [] +327ms

Connected Devices:

No connected devices found

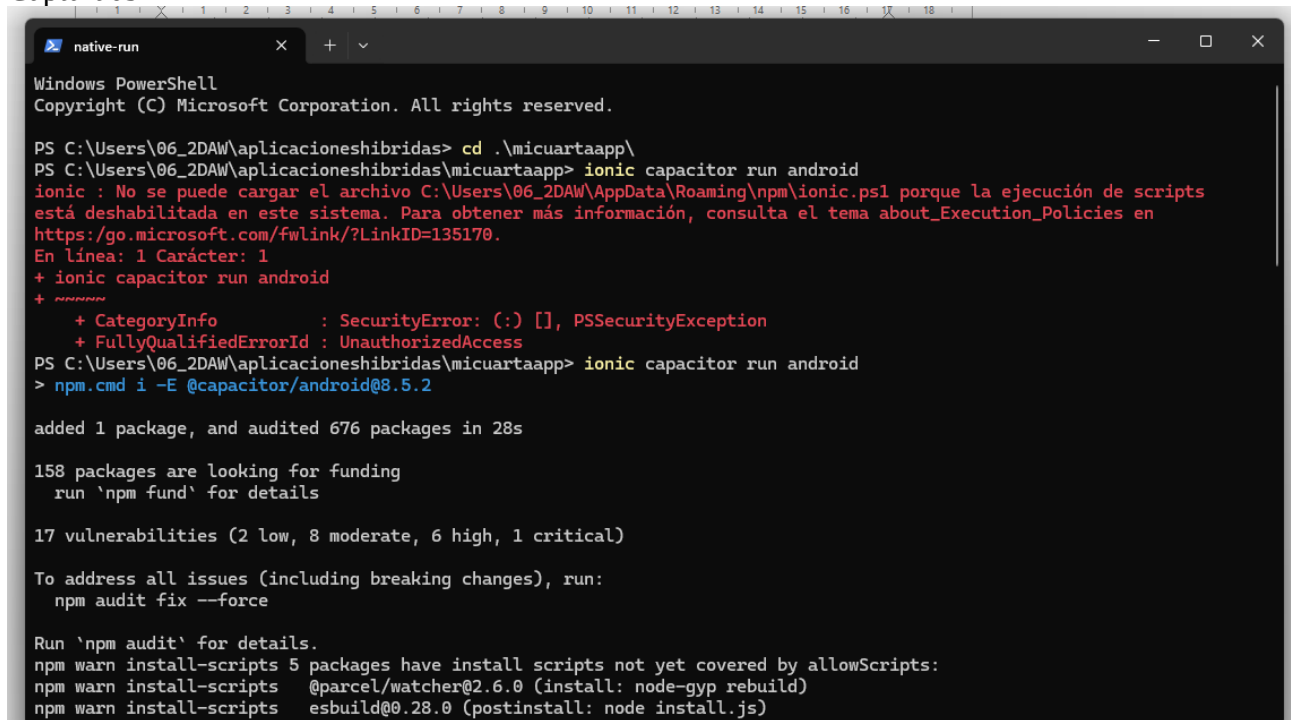
Virtual Devices:

  Name      API      Target ID
  -----
  Pixel 7   API 37.2  Pixel_7

C:\Users\06_2DAW\aplicacioneshibridas\mi-primer-app>
```

Desde la carpeta del proyecto se ejecuta `npx native-run android --list --verbose`: no se detectan dispositivos físicos conectados, pero sí el dispositivo virtual Pixel 7 disponible para lanzar la app.

Captura09



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\06_2DAW\aplicacioneshibridas> cd .\micuartaapp\
PS C:\Users\06_2DAW\aplicacioneshibridas\micuartaapp> ionic capacitor run android
ionic : No se puede cargar el archivo C:\Users\06_2DAW\AppData\Roaming\npm\ionic.ps1 porque la ejecución de scripts
está deshabilitada en este sistema. Para obtener más información, consulta el tema about_Execution_Policies en
https://go.microsoft.com/fwlink/?LinkID=135170.
En línea: 1 Carácter: 1
+ ionic capacitor run android
+ ~~~~~
+ CategoryInfo          : SecurityError: (:) [], PSSecurityException
+ FullyQualifiedErrorId : UnauthorizedAccess

PS C:\Users\06_2DAW\aplicacioneshibridas\micuartaapp> ionic capacitor run android
> npm.cmd i -E @capacitor/android@8.5.2

added 1 package, and audited 676 packages in 28s

158 packages are looking for funding
  run `npm fund` for details

17 vulnerabilities (2 low, 8 moderate, 6 high, 1 critical)

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.
npm warn install-scripts 5 packages have install scripts not yet covered by allowScripts:
npm warn install-scripts @parcel/watcher@2.6.0 (install: node-gyp rebuild)
npm warn install-scripts esbuild@0.28.0 (postinstall: node install.js)
```

Incidencia: ejecución de scripts bloqueada en PowerShell

Al ejecutar "ionic capacitor run android" en Windows PowerShell, apareció el error "No se puede cargar el archivo ionic.ps1 porque la ejecución de scripts está deshabilitada en este sistema" (PSSecurityException, UnauthorizedAccess).

Causa: Ionic CLI se instala con npm como un script de PowerShell (ionic.ps1), y la política de ejecución por defecto de Windows impide ejecutar scripts. No es un fallo de Ionic ni del proyecto.

Solución: se cambió la política de ejecución solo para el usuario actual, sin permisos de administrador, con el comando: Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned

Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned

Resultado: tras reiniciar la terminal, el comando "ionic capacitor run android" se ejecutó correctamente.

captura10

```
Windows PowerShell
chunk-0FfhY9iF.js | p-DSuBagxs | 10.06 kB | 2.59 kB
chunk-v8hl1AbD.js | p-DcrAT6v8 | 4.77 kB | 1.93 kB
chunk-BpDXp-HY.js | home-page | 4.01 kB | 1.73 kB
chunk-CUXoqvLT.js | view-message-page | 2.69 kB | 1.30 kB
chunk-DsBqGRCX.js | - | 2.37 kB | 1.07 kB
chunk-DYu2-mnw.js | p-CcOL6jEv | 1.56 kB | 711 bytes
chunk-BvcbF8Ef.js | p-BmVRXR1y | 932 bytes | 932 bytes
chunk-CZjwaLUQ.js | p-CDvyV0r0 | 879 bytes | 879 bytes
chunk-BCWthadI.js | p-S-gOI4A | 592 bytes | 592 bytes
chunk-Ce1x29z8.js | p-CpcdCRug | 412 bytes | 412 bytes

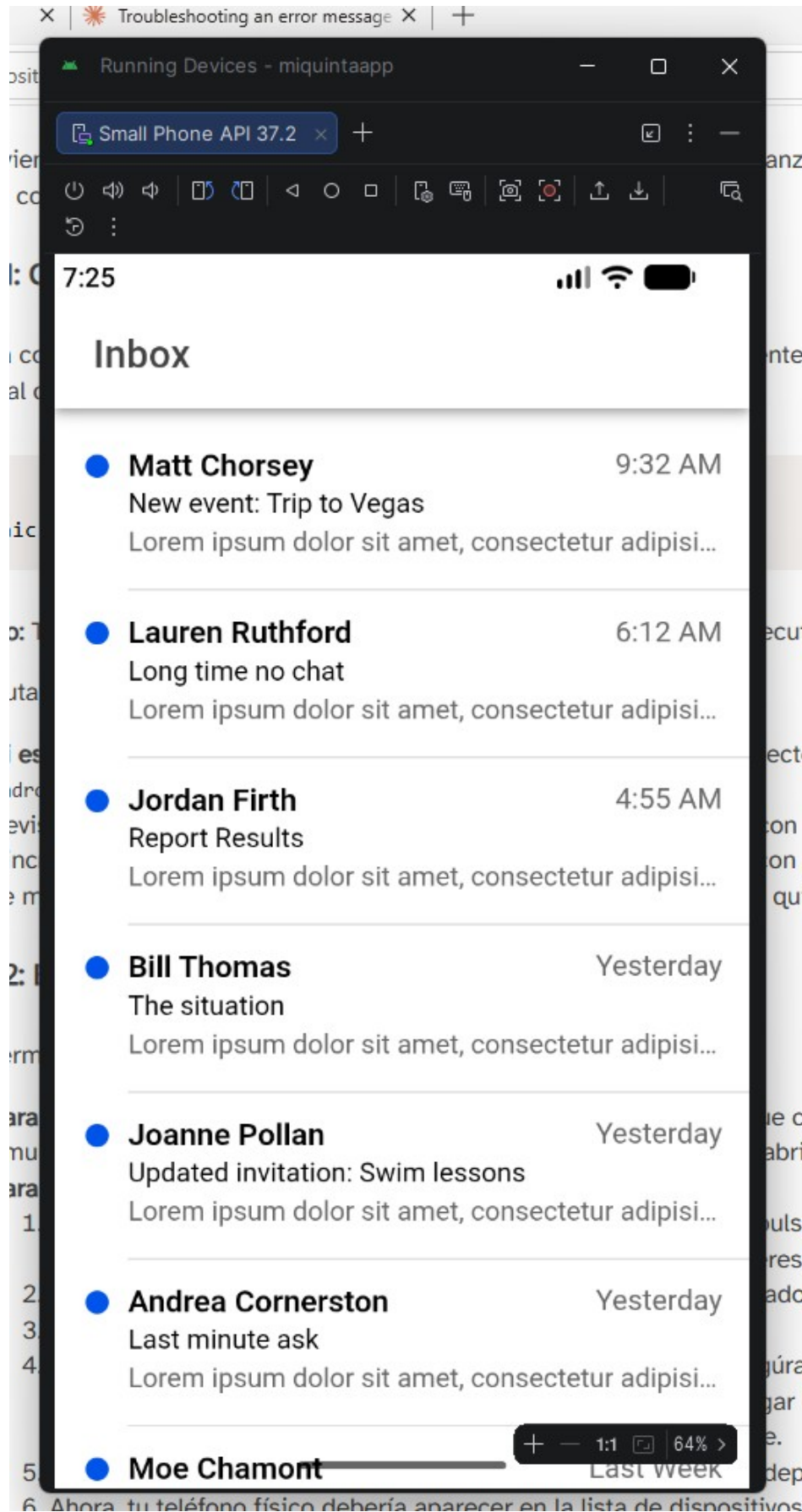
Application bundle generation complete. [21.480 seconds] - 2026-10-02T07:09:28.631Z

Output location: C:\Users\06_2DAW\aplicacioneshibridas\micuartaapp\www

> capacitor.cmd run android --target Small_Phone
[capacitor] / Copying web assets from www to android\app\src\main\assets\public in 16.51ms
[capacitor] / Creating capacitor.config.json in android\app\src\main\assets in 1.21ms
[capacitor] [info] Inlining sourcemaps
[capacitor] / copy android in 45.70ms
[capacitor] / Updating Android plugins in 4.74ms
[capacitor] [info] Found 4 Capacitor plugins for android:
[capacitor] @capacitor/app@8.1.1
[capacitor] @capacitor/haptics@8.0.2
[capacitor] @capacitor/keyboard@8.0.5
[capacitor] @capacitor/status-bar@8.0.3
[capacitor] / update android in 69.87ms
[capacitor] / Running Gradle build in 752.79s
[capacitor] / Deploying app-debug.apk to Small_Phone in 4.13s
PS C:\Users\06_2DAW\aplicacioneshibridas\micuartaapp> ionic capacitor run android|
```

Aplicación ejecutada de manera exitosa

Captura 11



Directo de la aplicación ejecutada

The screenshot displays a development environment with three main components:

- Code Editor (Claude):** Shows a file explorer on the left with a project named "mi-quintaapp". The main editor area displays a table of file sizes for various assets.
- Terminal (Windows PowerShell):** Shows the output of the command `capacitor.cmd run android --target Small_Phone`. The output indicates that the application bundle generation is complete and provides the output location: `C:\Users\06_2DAW\aplicacioneshibridas\mi-primer-app\www`.
- Android Emulator (Small Phone):** Shows a mobile interface with a status bar at the top displaying the time 8:07. The main content area displays "Tab 1" and "Hola mundo". Below this, there is a paragraph of text in Spanish: "Esta es mi primera aplicación híbrida hecha en Ionic". At the bottom, there is a navigation bar with three tabs: "Tab 1", "Tab 2", and "Tab 3".

The terminal output also includes the following information:

- Application bundle generation complete. [18.857 seconds] - 2026-10-02T08:03:52.613Z
- Output location: C:\Users\06_2DAW\aplicacioneshibridas\mi-primer-app\www
- Command: `> capacitor.cmd run android --target Small_Phone`
- Log messages: `[capacitor] / Copying web assets from www to android/app/src/main/assets/public in 20.91ms`, `[capacitor] / Creating capacitor.config.json in android/app/src/main/assets in 1.60ms`, `[capacitor] [info] Inlining source maps`, `[capacitor] / copy android in 57.01ms`, `[capacitor] / Updating Android plugins in 6.01ms`, `[capacitor] [info] Found 4 Capacitor plugins for android:`, `[capacitor] @capacitor/app@1.1`, `[capacitor] @capacitor/haptics@8.0.2`, `[capacitor] @capacitor/keyboard@8.0.5`, `[capacitor] @capacitor/status-bar@8.0.3`, `[capacitor] / update android in 79.25ms`, `[capacitor] / Running Gradle build in 14.39s`, `[capacitor] / Deploying app-debug.apk to Small_Phone in 3.72s`

The code editor also shows a "Project Structure" section with the following files:

- android
- ios
- src
- www

